

Finding Small Steps for Hard Questions, in work on the Process Semantics of Regular Expressions

<https://clegra.github.io>

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Department of Computer Science

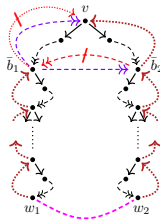
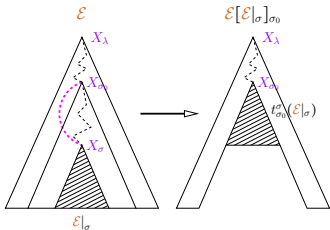


L'Aquila, Italy

IFIP 1.6 Working Group Meeting

Lisbon

July 18, 2026



Overview

- ▶ Regular expressions (also: 1-free/under-star-1-free $(*/\perp)$ expr.s)
- ▶ Milner's process interpretation P /semantics $\llbracket \cdot \rrbracket_P$
 - ▶ P -/ $\llbracket \cdot \rrbracket_P$ -expressible graphs ($\leadsto$ expressibility questions)
 - ▶ axioms for $\llbracket \cdot \rrbracket_P$ -identity ($\leadsto$ completeness question)
- ▶ **Small steps** for **hard questions**
 - (st1) reduction steps for well-behaved graphs under bisimilarity
 - ▶ $\llbracket \cdot \rrbracket_P$ -expressibility is decidable
 - (st2) collapse steps for LLEE graphs under bisimilarity
 - ▶ completeness for $(*/\perp)$ expressions
 - (st3) collapse steps for LLEE 1-graphs under bisimilarity, as far as possible
 - ▶ counterexample to collapse, but leads to completeness
- ▶ Interpretation–readback **correspondences**
 - (irc) LLEE/1-LLEE graphs $\triangleq$ image of $P^\bullet$ restricted to $(*/\perp)$ /all expr.s
 - ▶ where $P^\bullet$ a compact refinement of P
 - ▶ crucial for P -/ $\llbracket \cdot \rrbracket_P$ -expressibility, and completeness questions
- ▶ Outlook

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Regular Expression and Milner's Process Interpretation

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Process interpretation P of regular expressions *(Milner, 1984)*

$0 \xrightarrow{P} \text{deadlock } \delta, \text{ no termination}$

$a \xrightarrow{P} \text{atomic action } a, \text{ then terminate}$

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$e_1 \cdot e_2 \xrightarrow{P} (\text{sequentialization}) \text{ execute } P(e_1), \text{ then } P(e_2)$

$e^* \xrightarrow{P} (\text{iteration}) \text{ repeat (terminate or execute } P(e))$

Process interpretation P of regular expressions (Milner, 1984)

$0 \xrightarrow{P}$ deadlock δ , no termination

$0^* \xrightarrow{P}$ empty-step process ϵ , then terminate

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Process semantics $\llbracket \cdot \rrbracket^P$ of regular expressions *(Milner, 1984)*

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$e^* \xrightarrow{P}$ (*iteration*) repeat (terminate or execute $P(e)$)

$\llbracket e \rrbracket_P := \llbracket P(e) \rrbracket_{\leftrightarrow}$ (*bisimilarity* equivalence class of process $P(e)$)

Regular Expressions

Definition (~ *Copi-Elgot-Wright, 1958*)

Regular expressions over alphabet A with unary Kleene star:

$e, e_1, e_2 ::= 0 \mid a \mid e_1 + e_2 \mid e_1 \cdot e_2 \mid e^*$ (for $a \in A$).

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- ▶ symbol **0** instead of $\emptyset$, symbol **1** instead of $\{\epsilon\}$
- ▶ with unary star $*$: **1** is definable as **0***

Regular Expressions

Definition (*~ Kleene, 1951, ~ Copi-Elgot-Wright, 1958*)

Regular expressions over alphabet A with unary / binary Kleene star:

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$$e, e_1, e_2 ::= 0 \mid 1 \mid a \mid e_1 + e_2 \mid e_1 \cdot e_2 \mid e_1^{\oplus} e_2 \quad (\text{for } a \in A).$$

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Regular Expressions (1-free)

Definition (*~ Kleene, 1951, ~ Copi-Elgot-Wright, 1958*)

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Definition (for process interpretation)

1-free regular expressions over alphabet A with binary Kleene star:

$$f, f_1, f_2 ::= 0 \mid a \mid f_1 + f_2 \mid f_1 \cdot f_2 \mid f_1^{\oplus} f_2 \quad (\text{for } a \in A).$$

Regular Expressions (1-free)

Definition (*~ Kleene, 1951, ~ Copi-Elgot-Wright, 1958*)

Regular expressions over alphabet A with unary / binary Kleene star:

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Definition (for process interpretation)

1-free regular expressions over alphabet A with unary / binary Kleene star:

$$f, f_1, f_2 ::= 0 \mid a \mid f_1 + f_2 \mid f_1 \cdot f_2 \mid (f_1^*) \cdot f_2 \quad (\text{for } a \in A),$$

$$f, f_1, f_2 ::= 0 \mid a \mid f_1 + f_2 \mid f_1 \cdot f_2 \mid f_1^{\otimes} f_2 \quad (\text{for } a \in A).$$

Regular Expressions (under-star-/1-free)

Definition ($\sim$ Kleene, 1951, $\sim$ Copi-Elgot-Wright, 1958)

Regular expressions over alphabet A with unary / binary Kleene star:

$$e, e_1, e_2 ::= \mathbf{0} \mid \mathbf{1} \mid \mathbf{a} \mid e_1 + e_2 \mid e_1 \cdot e_2 \mid e^* \quad (\text{for } \mathbf{a} \in A).$$

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- ▶ with binary star $\otimes$: $\mathbf{1}$ is **not** definable (in its absence)

Definition (for process interpretation)

The set $\text{RExp}^{(+)}(A)$ of **1-free regular expressions** over A is defined by:

$$f, f_1, f_2 ::=_{\text{assoc}(\cdot)} \mathbf{0} \mid \mathbf{a} \mid f_1 + f_2 \mid f_1 \cdot f_2 \mid f_1^* \cdot f_2 \quad (\text{for } \mathbf{a} \in A),$$

the set $\text{RExp}^{(*/+)}(A)$ of **under-star-1-free regular expressions** over A by:

$$uf, uf_1, uf_2 ::= \mathbf{0} \mid \mathbf{1} \mid \mathbf{a} \mid uf_1 + uf_2 \mid uf_1 \cdot uf_2 \mid f^* \quad (\text{for } \mathbf{a} \in A).$$

Regular Expressions $((*/\perp)/1\text{-free})$

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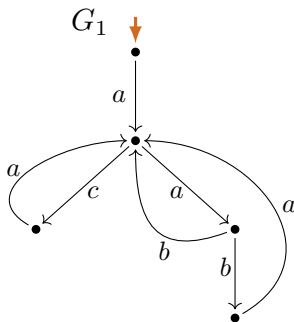
The set $\text{RExp}^{(+)}(A)$ of $\mathbf{1}$ -free regular expressions over A is defined by:

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the set $\text{RExp}^{(*, \perp)}(A)$ of under-star-1-free $((*/\perp)$ reg. expr. over A by:

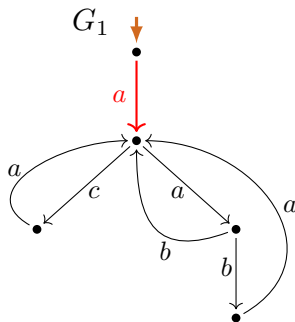
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P -expressibility and $[[\cdot]]_P$ -expressibility (example, informally)



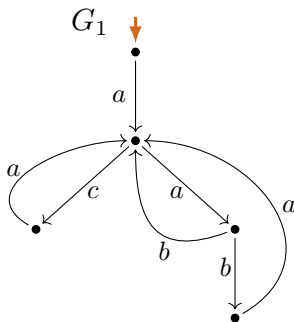
$$P\left(\overbrace{a \cdot (c \cdot a + a \cdot (b + b \cdot a))^* \cdot 0}^f \right)$$

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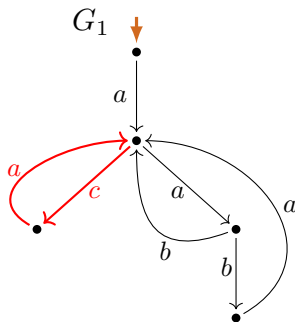
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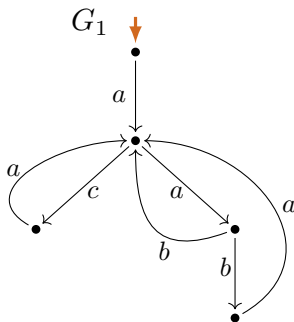
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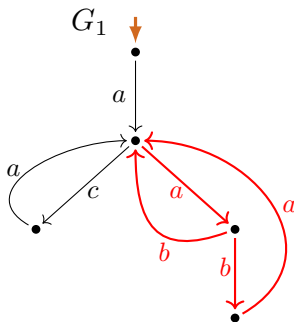
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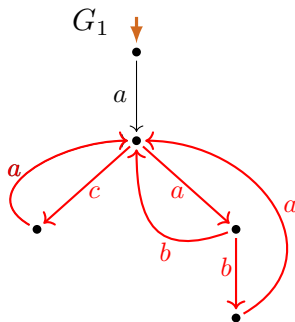
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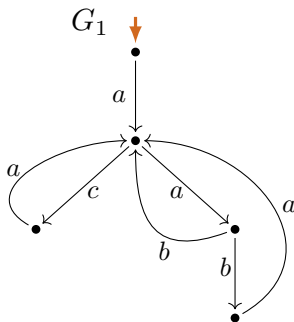
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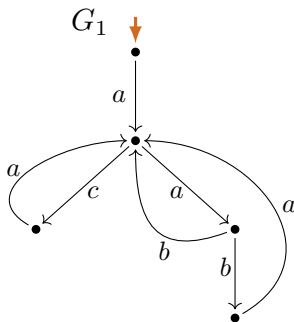
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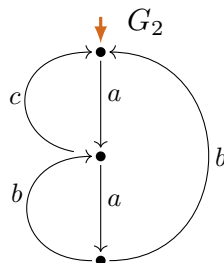
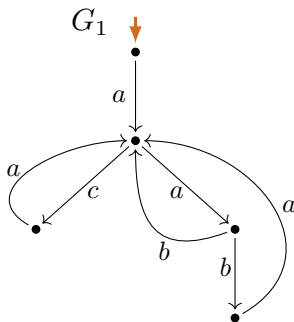
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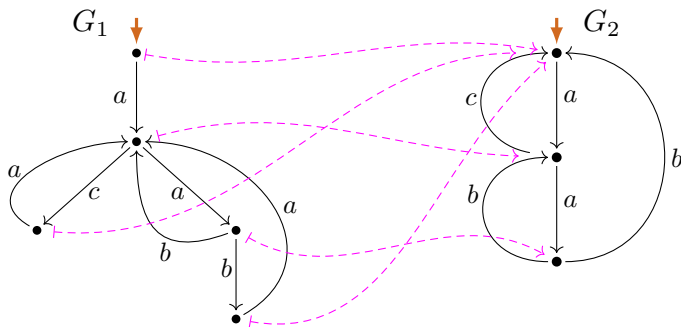
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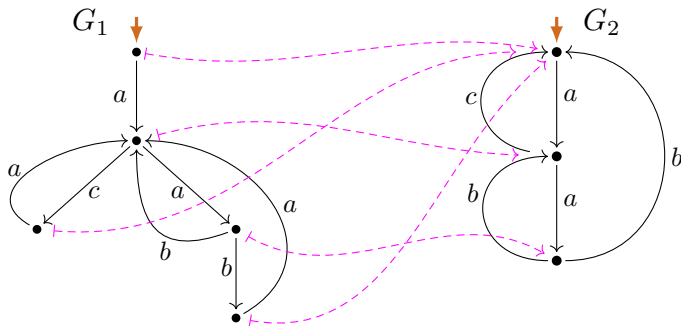
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Process interpretation P (formally)

Definition (Transition system specification $\mathcal{T}$)

$$\frac{}{a \xrightarrow{a} 1} \quad \frac{e_i \xrightarrow{a} e'_i}{e_1 + e_2 \xrightarrow{a} e'_i} \quad (i \in \{1, 2\})$$

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Process interpretation P (formally)

Definition (Transition system specification $\mathcal{T}$)

$$\begin{array}{c}
 \frac{}{1 \Downarrow} \qquad \frac{e_i \Downarrow}{(e_1 + e_2) \Downarrow} \ (i \in \{1, 2\}) \qquad \frac{e_1 \Downarrow \quad e_2 \Downarrow}{(e_1 \cdot e_2) \Downarrow} \qquad \frac{}{(e^*) \Downarrow} \\
 \\
 \frac{}{a \xrightarrow{a} 1} \qquad \frac{e_i \xrightarrow{a} e'_i}{e_1 + e_2 \xrightarrow{a} e'_i} \ (i \in \{1, 2\}) \\
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 \frac{}{a \xrightarrow{a} 1} \quad \frac{e_i \xrightarrow{a} e'_i}{e_1 + e_2 \xrightarrow{a} e'_i} \ (i \in \{1, 2\}) \\
 \\
 \frac{e_1 \xrightarrow{a} e'_1}{e_1 \cdot e_2 \xrightarrow{a} e'_1 \cdot e_2} \quad \frac{e_1 \Downarrow \quad e_2 \xrightarrow{a} e'_2}{e_1 \cdot e_2 \xrightarrow{a} e'_2} \quad \frac{e \xrightarrow{a} e'}{e^* \xrightarrow{a} e' \cdot e^*}
 \end{array}$$

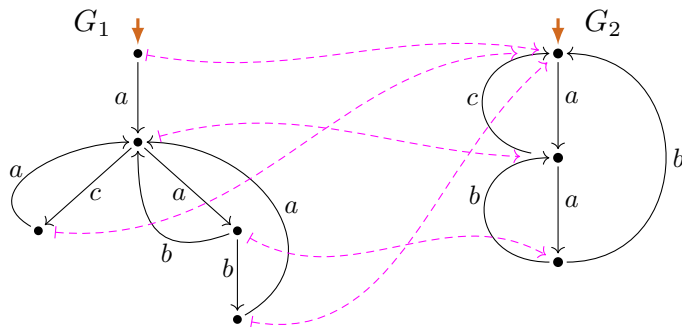
Definition

The **process interpretation** $P(e)$ of a regular expression e :

$P(e) :=$ **labeled transition graph** generated by e by derivations in $\mathcal{T}$,

The **process semantics** $\llbracket e \rrbracket_P$ of e is: $\llbracket e \rrbracket_P := [P(e)]_{\leftrightarrow}$.

P -expressibility and $\llbracket \cdot \rrbracket_P$ -expressibility (example, informally)

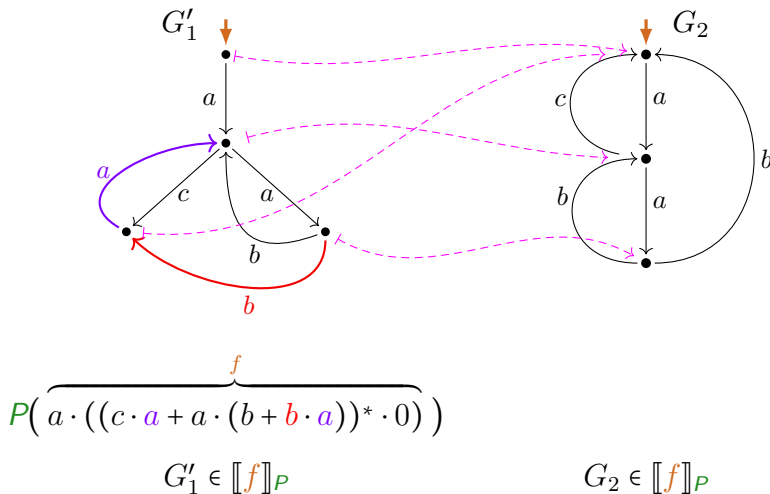


$$P\left(\overbrace{a \cdot ((c \cdot a + a \cdot (b + b \cdot a))^* \cdot 0)}^f\right)$$

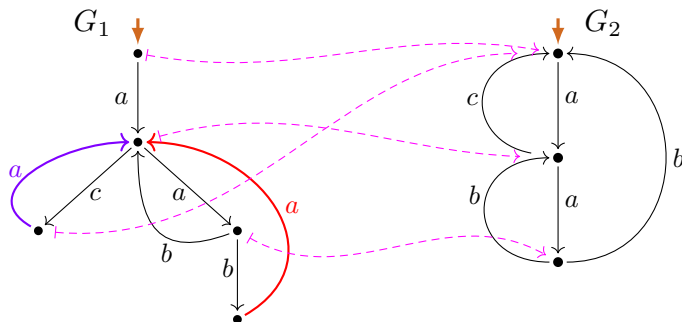
$$G_1 \in \llbracket f \rrbracket_P$$

$$G_2 \in \llbracket f \rrbracket_P$$

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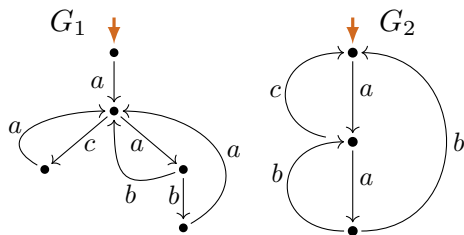


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P -expressibility and $\llbracket \cdot \rrbracket_P$ -expressibility (examples)

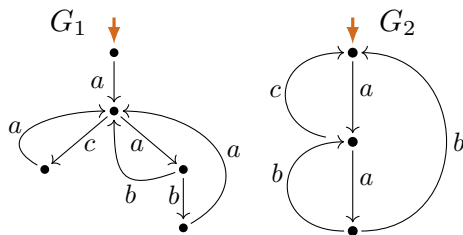


P -expressible

$\llbracket \cdot \rrbracket_P$ -expressible

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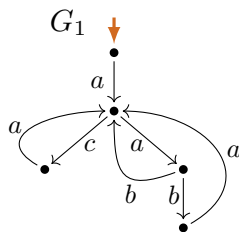
P -expressible

?

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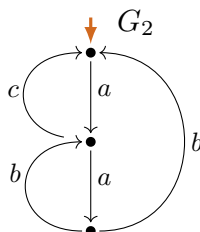
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P -expressibility and $\llbracket \cdot \rrbracket_P$ -expressibility (examples)



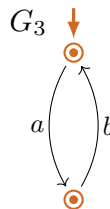
P -expressible

$\llbracket \cdot \rrbracket_P$ -expressible



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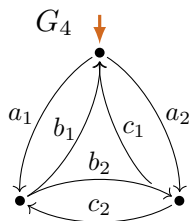
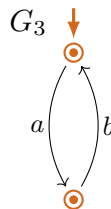
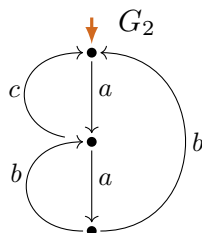
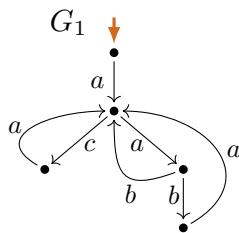
$\llbracket \cdot \rrbracket_P$ -expressible



not P -expressible

not $\llbracket \cdot \rrbracket_P$ -expressible

P -expressibility and $\llbracket \cdot \rrbracket_P$ -expressibility (examples)



P -expressible

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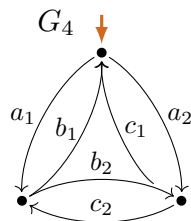
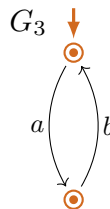
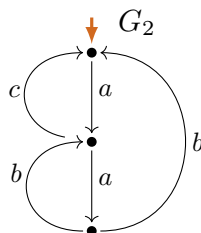
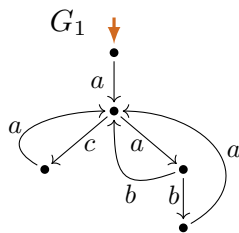
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Expressibility Questions:

How can P -expressibility and $\llbracket \cdot \rrbracket_P$ -expressibility be characterized?

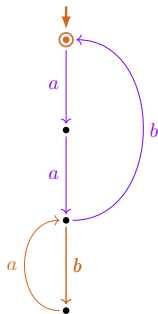
Overview

- ▶ Regular expressions (also: 1-free/under-star-1-free $(*/\perp)$ expr.s)
- ▶ Milner's process interpretation P /semantics $\llbracket \cdot \rrbracket_P$
 - ▶ P -/ $\llbracket \cdot \rrbracket_P$ -expressible graphs ($\leadsto$ expressibility questions)
 - ▶ axioms for $\llbracket \cdot \rrbracket_P$ -identity ($\leadsto$ completeness question)
- ▶ **Small steps** for **hard questions**
 - (st1) **reduction steps** for **well-behaved** graphs under bisimilarity
 - ▶ $\llbracket \cdot \rrbracket_P$ -**expressibility** is decidable
 - (st2) **collapse steps** for **LLEE** graphs under bisimilarity
 - ▶ **completeness** for $(*/\perp)$ expressions
 - (st3) **collapse steps** for **LLEE 1**-graphs under bisimilarity, **as far as possible**
 - ▶ counterexample to collapse, but leads to **completeness**
- ▶ Interpretation–readback **correspondences**
 - (irc) **LLEE/1-LLEE** graphs $\triangleq$ image of $P^\bullet$ restricted to $(*/\perp)$ /all expr.s
 - ▶ where $P^\bullet$ a **compact** refinement of P
 - ▶ crucial for P -/ $\llbracket \cdot \rrbracket_P$ -**expressibility**, and **completeness** questions
- ▶ Outlook

Small steps for hard questions

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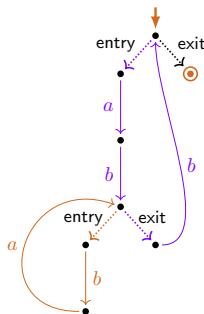
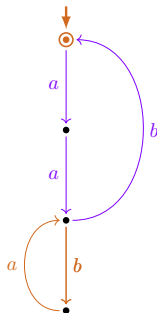
Well-behaved graphs



$$\sim P((aa(ba)^*b)^*)$$

$$P(1 \cdot (a \cdot ((a \cdot (b \cdot a)^*) \cdot b))^*)$$

Well-behaved graphs



well-behaved

loop-entry/-exit palm tree

(Corradini, Baeten)

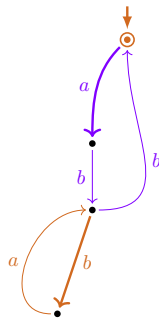
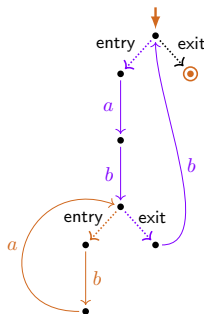
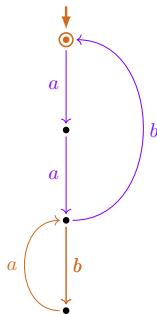
$$\sim P((aa(ba)^*b)^*)$$

$$\sim P((1 \cdot aa(1 \cdot ba)^* 1 \cdot b)^* (1 \cdot 1))$$

$$P(1 \cdot (a \cdot ((a \cdot (b \cdot a)^* \cdot b))^*))$$

$$P(1 \cdot (((1 \cdot a) \cdot (((a \cdot ((1 \cdot b) \cdot a)^* \cdot 1) \cdot b))^*) \cdot 1))$$

Well-behaved graphs



well-behaved
loop-entry/-exit palm tree
(Corradini, Baeten)

$$\sim P((aa(ba)^*b)^*)$$

$$\sim P(((1aa(1ba)^*1b)^*(1.1)))$$

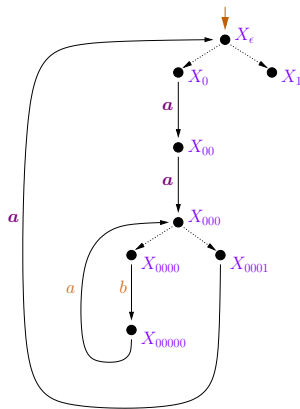
$$\sim P((aa(ba)^*b)^*)$$

$$P(1.(a.((a.(b.a)^*.b))^*)$$

$$P(1.((((1.a).(((a.(((1.b).a)^*.1).b))^*).1)))$$

$$P(1.(a.((a.(b.a)^*.b))^*)$$

Well-Behaved specification (example)



$$X_{\epsilon} = 1 \cdot X_0 + 1 \cdot X_1$$

$$X_0 = a \cdot X_{00}$$

$$X_{00} = a \cdot X_{000}$$

$$X_{000} = 1 \cdot X_{0000} + 1 \cdot X_{0001}$$

$$X_{0000} = b \cdot X_{00000}$$

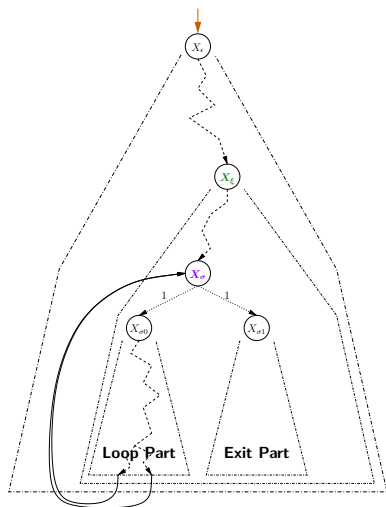
$$X_{00000} = a \cdot X_{000}$$

$$X_{0001} = a \cdot X_{\epsilon}$$

$$X_1 = 0$$

$$P((aa(ba)^*a)^*.0)$$

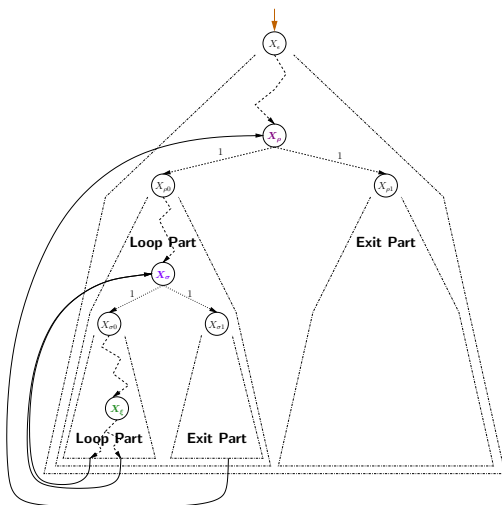
Well-Behaved specifications (well-beh./cycling variables)



$X_\xi, X_\epsilon \dots$ *well-behaved variables*
 (X_ξ “does not return” to a
 recursion variable above itself)

X_σ is a *cycling variable*
 (some recursion variable below X_σ
 “returns to” X_σ)

Well-Behaved specifications (cycles back to)



$X_\sigma, X_\rho \dots$ cycling variables

X_ξ cycles back to X_σ

(The nearest return of X_ξ
to a rec.var. above is to X_σ)

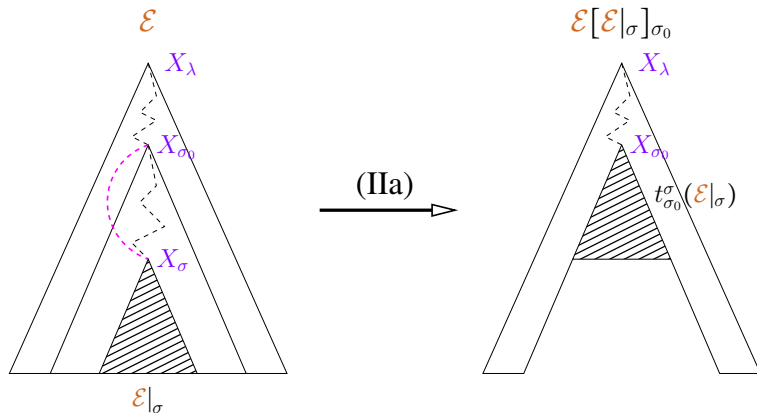
X_σ cycles back to X_ρ

$\llbracket \cdot \rrbracket_P$ -expressible = bisimilar to well-behaved

Theorem (Baeten/Corradini, 2005)

- (i) A graph G is $\llbracket \cdot \rrbracket_P$ -expressible
 $\iff G$ is the solution of a *well-behaved* specification.
- (ii) A graph G is $\llbracket \cdot \rrbracket_P$ -expressible
 $\iff$ there is a *well-behaved* graph G_{wb} such that $G \leftrightarrow G_{wb}$.

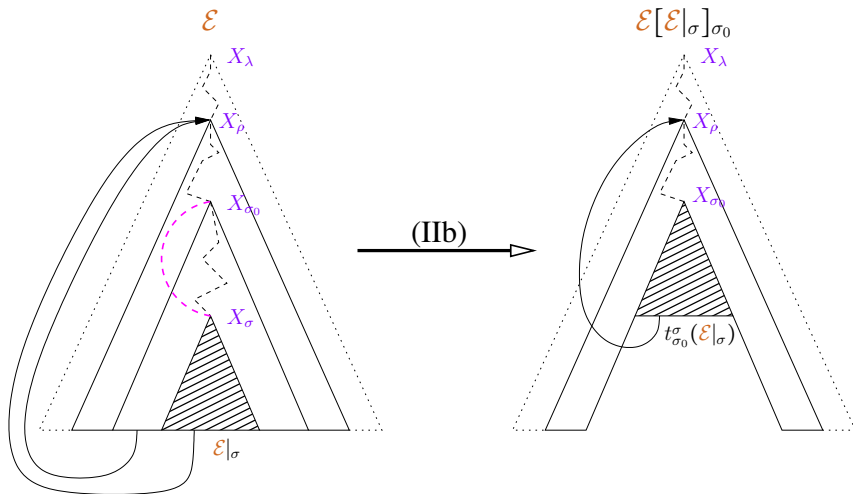
Reduction steps (IIa) for well-behaved graphs



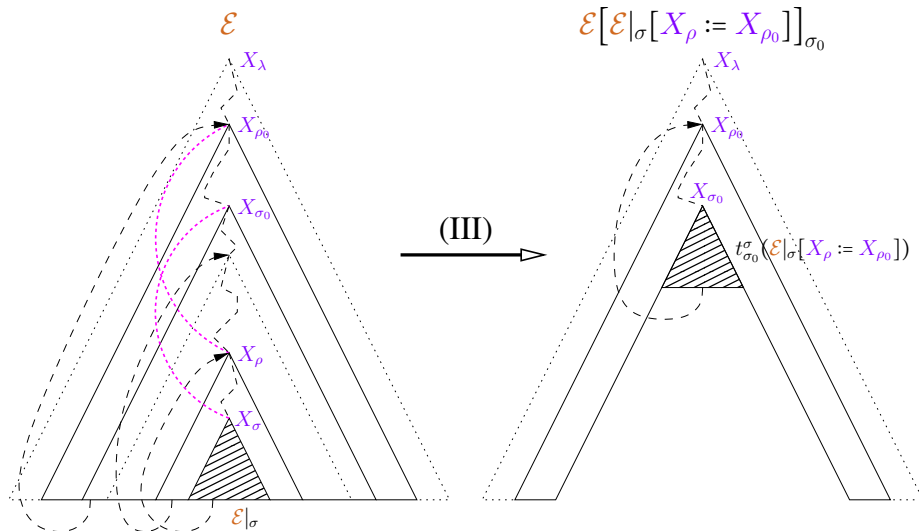
$$\langle X_{\sigma} | \mathcal{E} \rangle \Leftrightarrow \langle X_{\sigma_0} | \mathcal{E} \rangle$$

X_{σ}, X_{σ_0} are well-behaved

Reduction steps (IIb) for well-behaved graphs



Reduction steps (III) for well-behaved graphs



Result facilitated by steps (st1)

Theorem (*Baeten–Corradini–G, 2005*)

$\llbracket \cdot \rrbracket_P$ -Expressibility of graphs is *decidable* (at least super-exponentially).

Result facilitated by steps (st1)

Reduction Lemma for well-behaved graphs (G, 2005)

Let G be graph with n states and max. out-degree k such that $G \Leftrightarrow G_{wb}$ for a well-behaved graph G_{wb} (then G, G_{wb} are $\llbracket \cdot \rrbracket_P$ -expressible).

Then G_{wb} can be reduced under bisimilarity by reduction steps (I)–(III) to a bisimilar well-behaved graph $G_{wb}^{(0)}$ with:

- ▶ depth $\leq (n+1)^3 \cdot 2^{3k}$, and max. out-degree $\leq k$,
- ▶ $G_{wb}^{(0)} \Leftrightarrow G_{wb} (\Leftrightarrow G)$.

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Effective Expressibility Lemma (G, 2005)

A graph G with n states and max. out-degree k is $\llbracket \cdot \rrbracket_P$ -expressible

$\iff G \rightleftharpoons G_{wb}$ for a well-behaved graph G_{wb}
with depth $\leq (n+1)^3 \cdot 2^{3k}$ and max. out-degree $\leq k$.

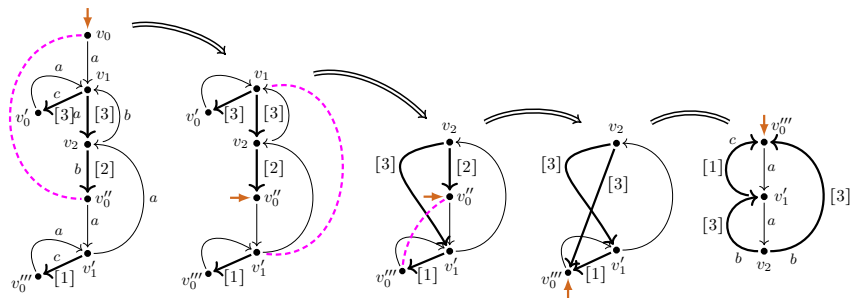
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Collapse steps for LLEE graphs under bisimilarity



Loop graphs (interpretations of innermost iterations without 1)

Definition

A process graph is a **loop graph** if:

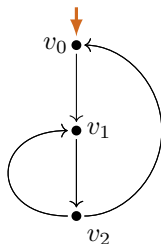
- (L1) There is an infinite path from the **start vertex**.
- (L2) Every infinite path from the **start vertex** returns to **it**.
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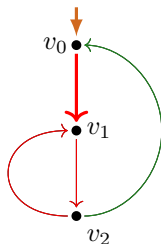


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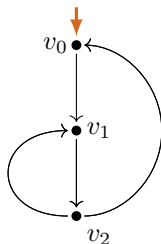
(L1), ~~(L2)~~

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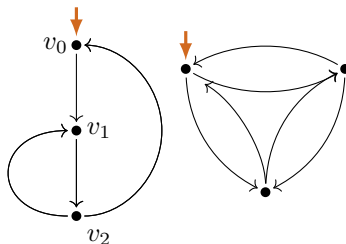
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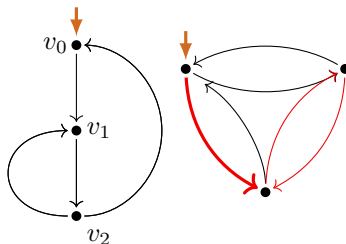
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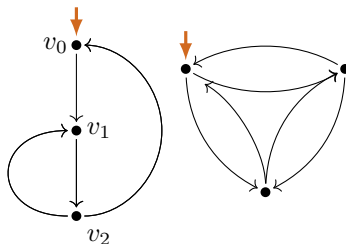
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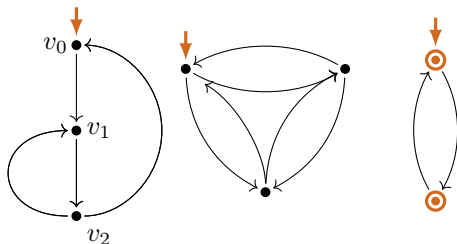
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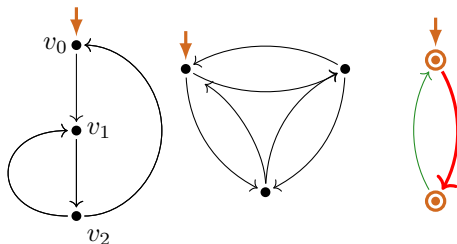
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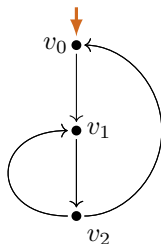
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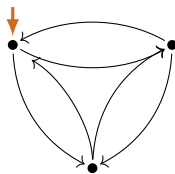
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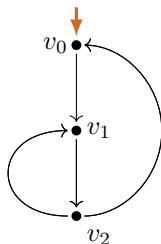


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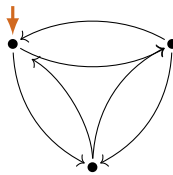
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(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~

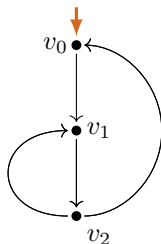


Loop graphs (interpretations of innermost iterations without 1)

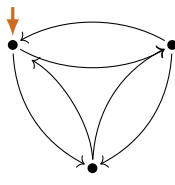
Definition

A process graph is a **loop graph** if:

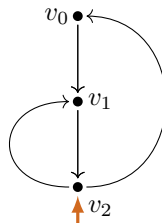
- (L1) There is an infinite path from the **start vertex**.
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- (L3) Immediate termination is **only** possible at the **start vertex**.



(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~

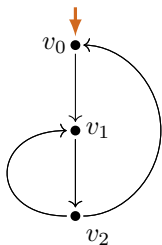


Loop graphs (interpretations of innermost iterations without 1)

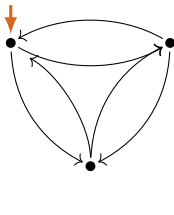
Definition

A process graph is a **loop graph** if:

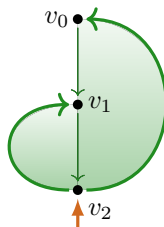
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(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~

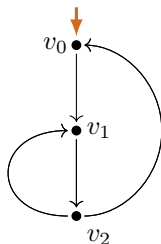


Loop graphs (interpretations of innermost iterations without 1)

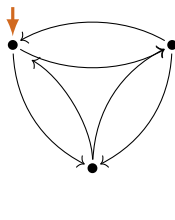
Definition

A process graph is a **loop graph** if:

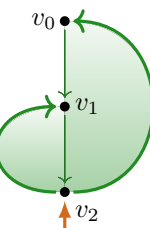
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(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



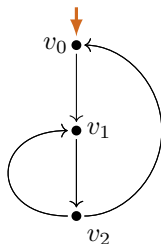
loop chart

Loop graphs (interpretations of innermost iterations without 1)

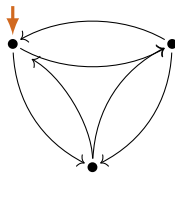
Definition

A process graph is a **loop graph** if:

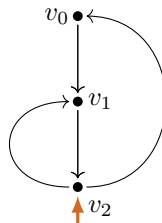
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(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



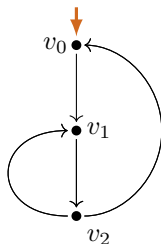
loop chart

Loop graphs (interpretations of innermost iterations without 1)

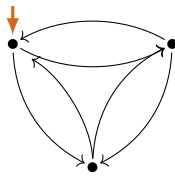
Definition

A process graph is a **loop graph** if:

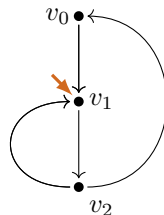
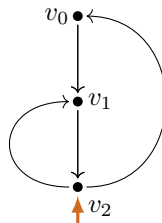
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(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~ **loop chart**

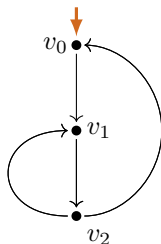


Loop graphs (interpretations of innermost iterations without 1)

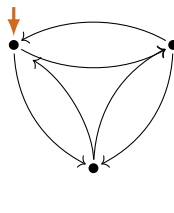
Definition

A process graph is a **loop graph** if:

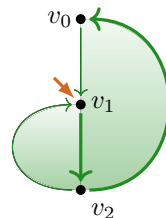
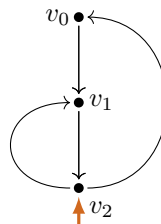
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(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~ loop chart

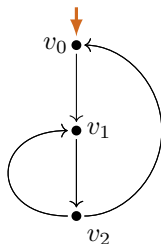


Loop graphs (interpretations of innermost iterations without 1)

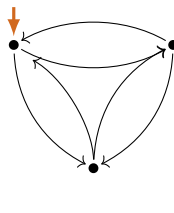
Definition

A process graph is a **loop graph** if:

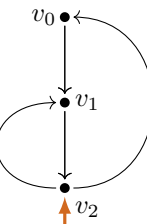
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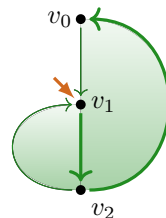
(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



loop chart



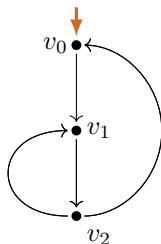
loop chart

Loop graphs (interpretations of innermost iterations without 1)

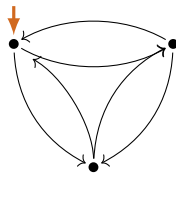
Definition

A process graph is a **loop graph** if:

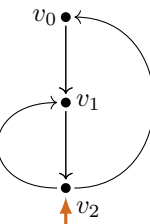
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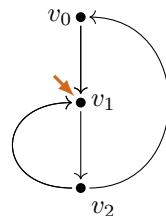
(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



loop chart



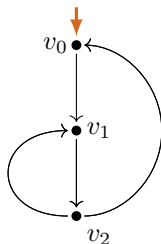
loop chart

Loop graphs (interpretations of innermost iterations without 1)

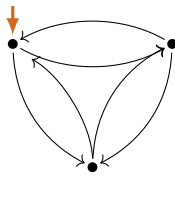
Definition

A process graph is a **loop graph** if:

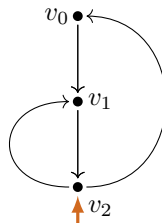
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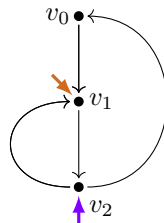
(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



loop chart

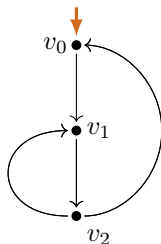


Loop graphs (interpretations of innermost iterations without 1)

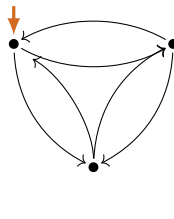
Definition

A process graph is a **loop graph** if:

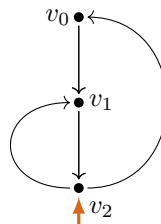
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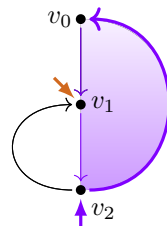
(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



loop chart

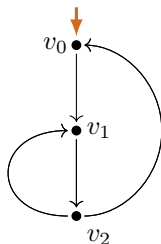


Loop graphs (interpretations of innermost iterations without 1)

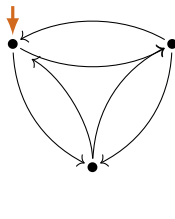
Definition

A process graph is a **loop graph** if:

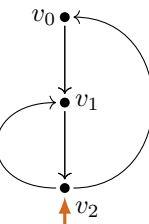
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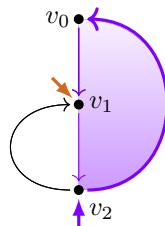
(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



loop chart



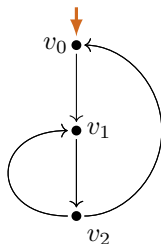
loop subchart

Loop graphs (interpretations of innermost iterations without 1)

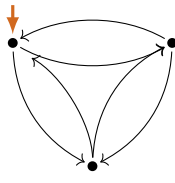
Definition

A process graph is a **loop graph** if:

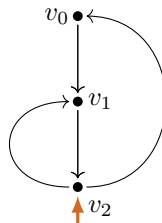
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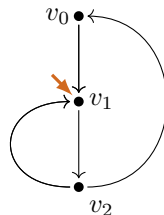
(L1), ~~(L2)~~



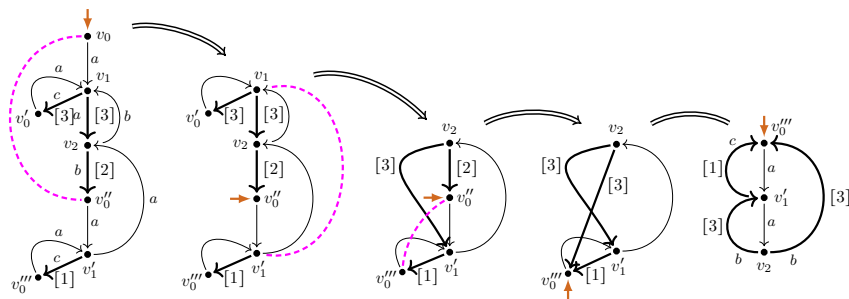
(L1), (L2), ~~(L3)~~



loop chart



Collapse steps for LLEE graphs under bisimilarity



L

Definition

A chart $\mathcal{C}$ satisfies **L** (*loop existence and elimination*) if:

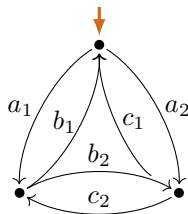
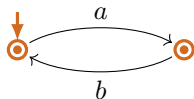
$$\begin{aligned}
 \exists \mathcal{C}_0 \big(& \mathcal{C} \xrightarrow[\text{elim}]{*} \mathcal{C}_0 \not\xrightarrow{\text{elim}} \\
 & \wedge \mathcal{C}_0 \text{ permits no infinite path} \big).
 \end{aligned}$$

L

Definition

A chart $\mathcal{C}$ satisfies **L**EE (*loop existence and elimination*) if:

$$\exists \mathcal{C}_0 \left(\mathcal{C} \xrightarrow{*}_{\text{elim}} \mathcal{C}_0 \not\rightarrow_{\text{elim}} \right. \\ \left. \wedge \mathcal{C}_0 \text{ permits no infinite path} \right).$$

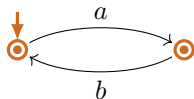


L

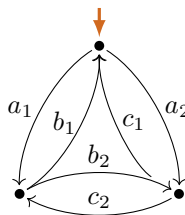
Definition

A chart $\mathcal{C}$ satisfies **L** (*loop existence and elimination*) if:

$$\exists \mathcal{C}_0 \left(\mathcal{C} \xrightarrow{*}_{\text{elim}} \mathcal{C}_0 \not\rightarrow_{\text{elim}} \right. \\ \left. \wedge \mathcal{C}_0 \text{ permits no infinite path} \right).$$

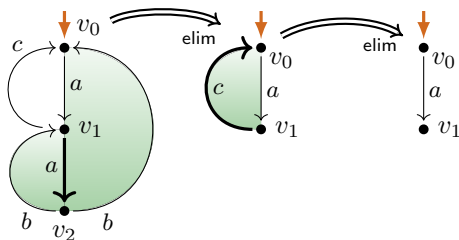


not L

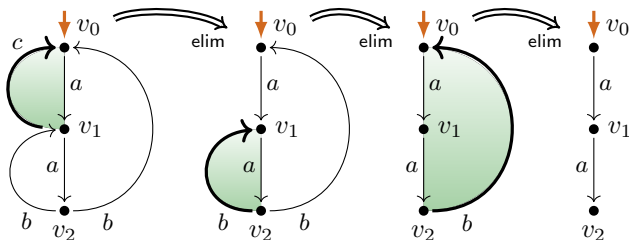
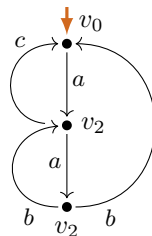


not L

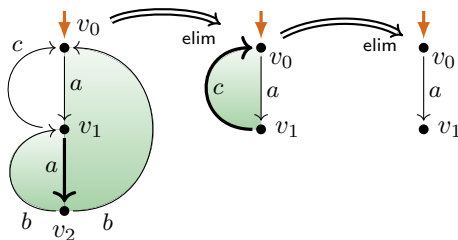
LEE



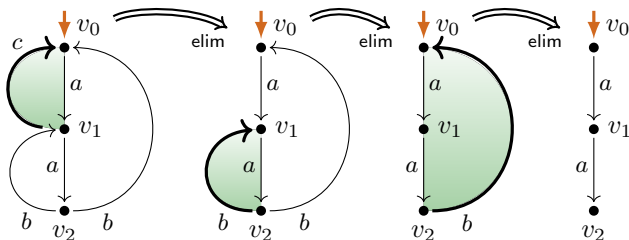
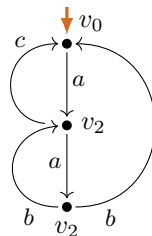
LEE



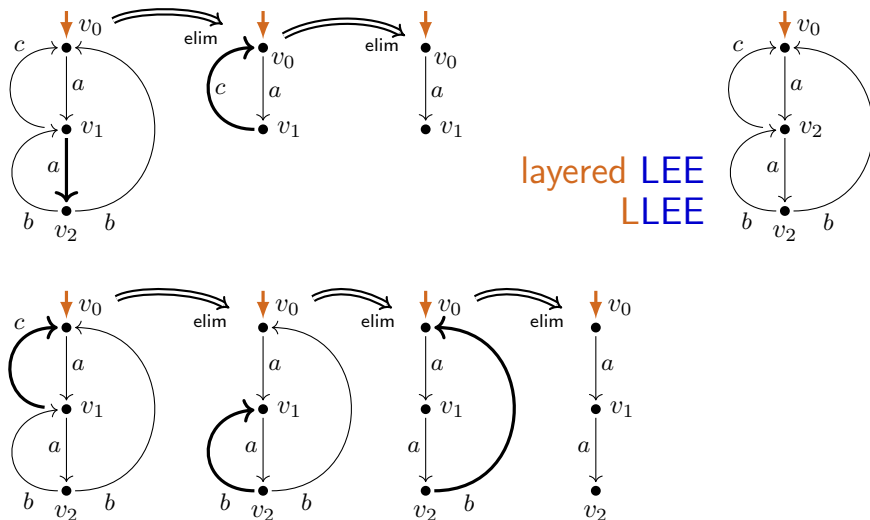
Layered LEE



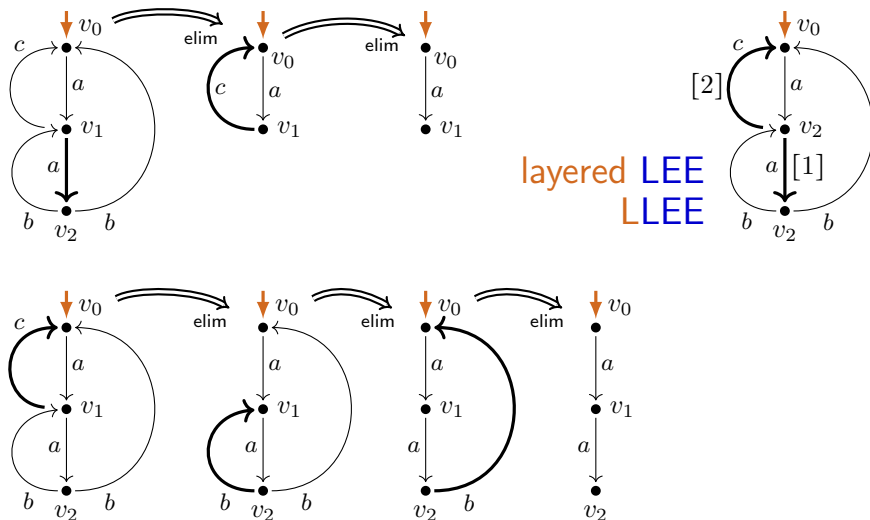
layered LEE
LLEE



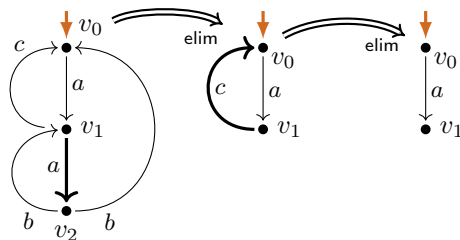
Layered LEE



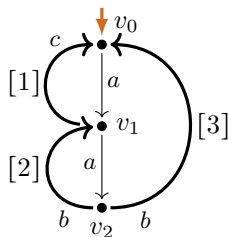
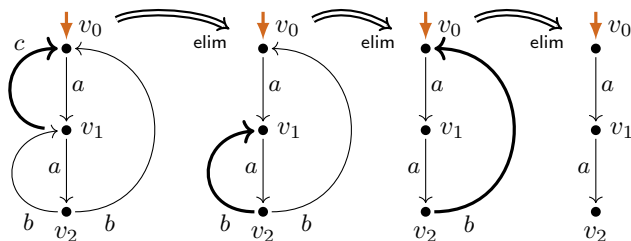
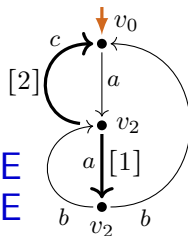
Layered LEE



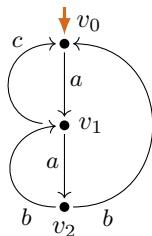
LLEE-witness and LLEE-graph



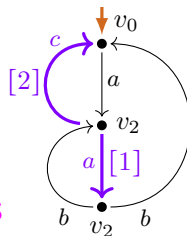
layered LEE
LLEE



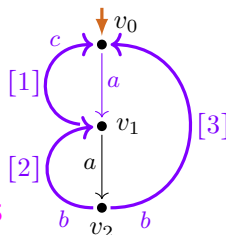
LLEE-witness and LLEE-graph



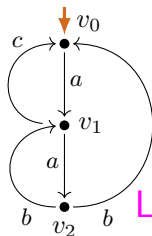
LLEE-witness



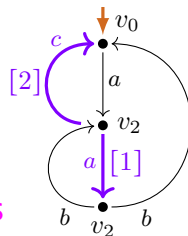
LLEE-witness



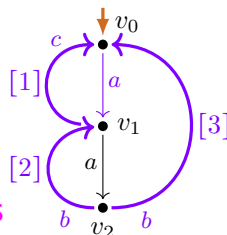
LLEE-witness and LLEE-graph



LLEE-graph

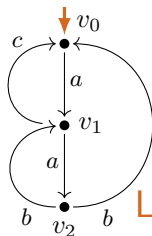


LLEE-witness



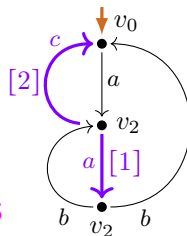
LLEE-witness

LLEE-witness and LLEE-graph

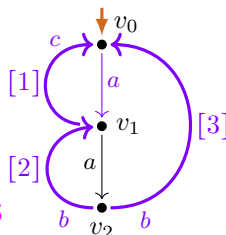


LLEE-graph

layered
LLEE-witness



layered
LLEE-witness



Properties of LEE-charts

Theorem ($\Leftarrow$ G/Fokkink, 2020)

A process graph G

is $\llbracket \cdot \rrbracket_P$ -expressible by an $(*/\perp)$ regular expression

(i.e. P -expressible modulo bisimilarity by an $(*/\perp)$ reg. expr.)

if and only if

the bisimulation collapse of G satisfies LEE.

Properties of LEE-charts

Theorem ($\Leftarrow$ *G/Fokkink, 2020*)

A process graph G

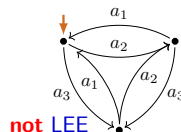
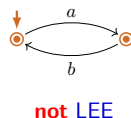
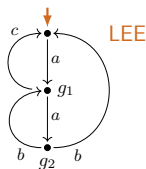
is $\llbracket \cdot \rrbracket_P$ -expressible by an $(*/\perp)$ regular expression

(i.e. P -expressible modulo bisimilarity by an $(*/\perp)$ reg. expr.)

if and only if

the bisimulation collapse of G satisfies LEE.

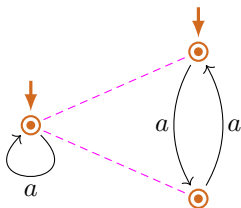
Hence $\llbracket \cdot \rrbracket_P$ -expressible | **not** $\llbracket \cdot \rrbracket_P$ -expressible by $(*/\perp)$ regular expressions:



L EE under bisimulation

Observation

- L EE is not invariant under bisimulation.



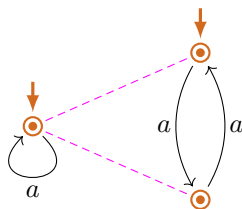
LEE

¬LEE

L EE under bisimulation

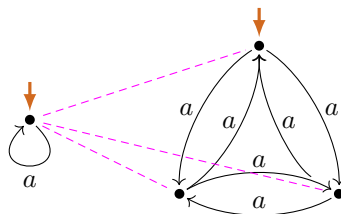
Observation

- L EE is **not** invariant under bisimulation.



L EE

¬L EE



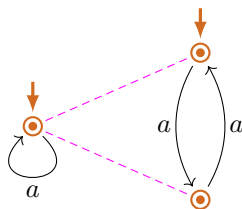
L EE

¬L EE

L EE under bisimulation

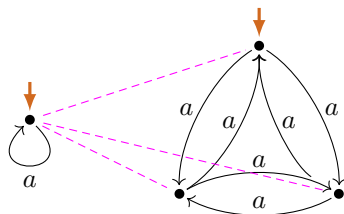
Observation

- ▶ L EE is **not** invariant under bisimulation.
- ▶ L EE is **not** preserved by converse functional bisimulation.



LEE

¬LEE



LEE

¬LEE

LEE under functional bisimulation

Lemma

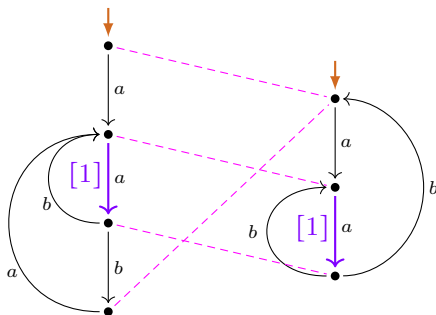
(i) LEE is preserved by *functional bisimulations*:

$$\text{LEE}(G_1) \wedge G_1 \rightsquigarrow G_2 \implies \text{LEE}(G_2).$$

Proof (Idea).

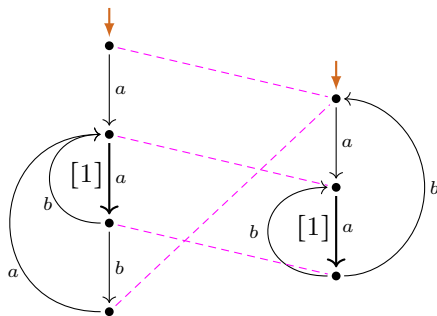
Use loop elimination in G_1 to carry out loop elimination in G_2 .

Collapsing LEE-witnesses

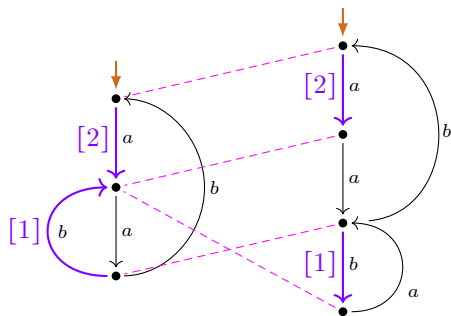


$$P(a(a(b + ba))^* \cdot 0)$$

Collapsing LEE-witnesses



$$P(a(a(b + ba))^* \cdot 0)$$



$$P((aa(ba))^* \cdot b)^* \cdot 0)$$

LEE under functional bisimulation / bisimulation collapse

Lemma

(i) LEE is preserved by *functional bisimulations*:

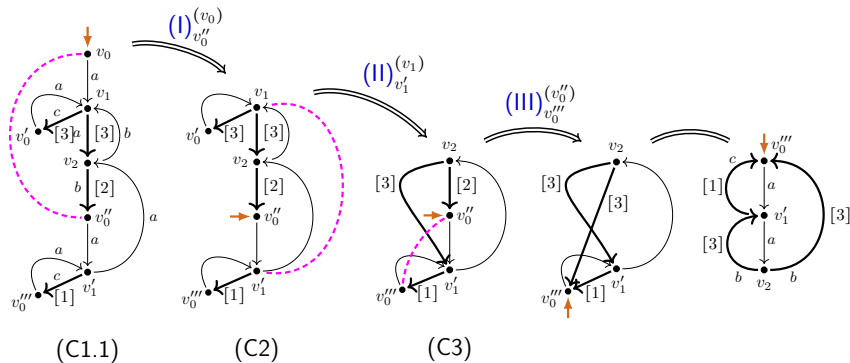
$$\text{LEE}(G_1) \wedge G_1 \rightrightarrows G_2 \implies \text{LEE}(G_2).$$

(ii) LEE is preserved from a process graph to its *bisimulation collapse*:

$$\text{LEE}(G) \wedge G \text{ has bisimulation collapse } C \implies \text{LEE}(C).$$

LLEE-preserving collapse of LLEE-charts (G/Fokkink, LICS'20)

(no 1-transitions!)

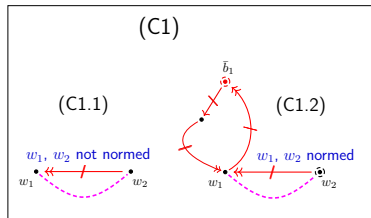


Lemma

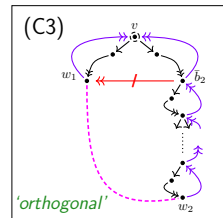
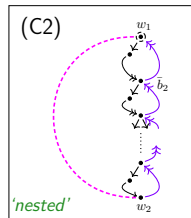
The bisimulation collapse of a LLEE-chart is again a LLEE-chart.

Reduced bisimilarity redundancies in LLEE-graphs (no 1-trans.!) (G/Fokkink, LICS'20)

w_1, w_2 in different scc's



w_1, w_2 in the same scc

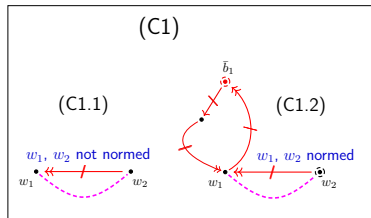


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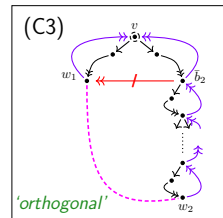
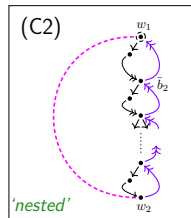
Every *not collapsed* LLEE-graph contains bisimilar vertices $w_1 \neq w_2$ of kind (C1), (C2), or (C3) (a *reduced bisimilarity redundancy* $\langle w_1, w_2 \rangle$):

Reduced bisimilarity redundancies in LLEE-graphs (no 1-trans.!) (G/Fokkink, LICS'20)

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Lemma

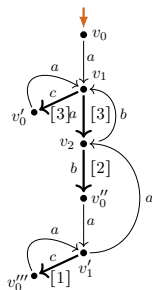
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Lemma

Every *reduced bisimilarity redundancy* in a LLEE-graph can be eliminated LLEE-preservingly.

LLEE-preserving collapse of LLEE-charts (G/Fokkink, LICS'20)

(no 1-transitions!)



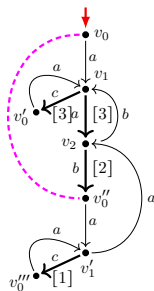
(C1.1)

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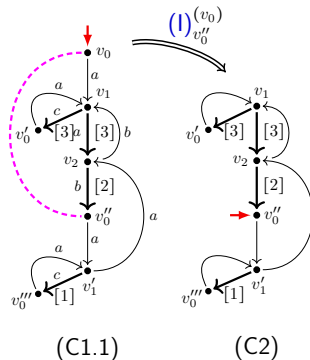
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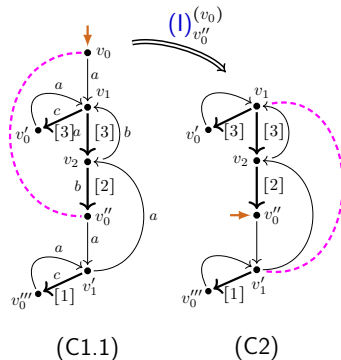


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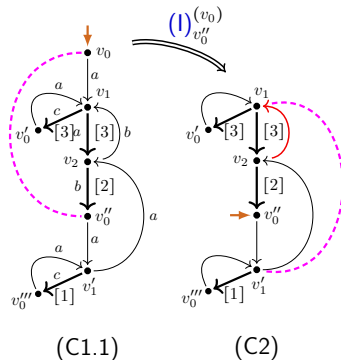


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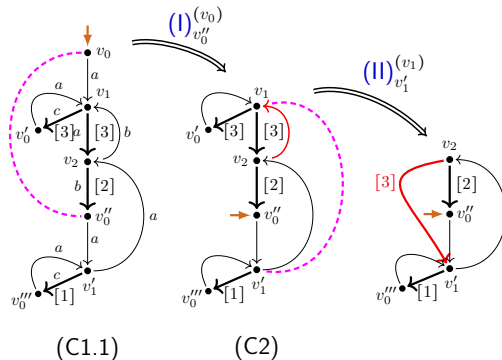


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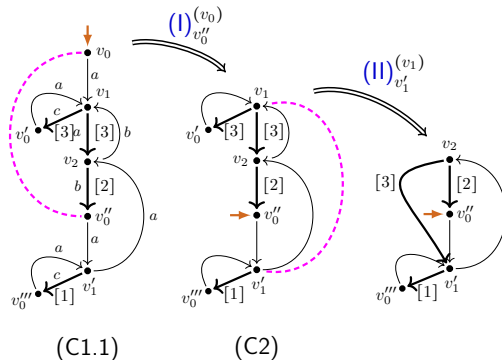


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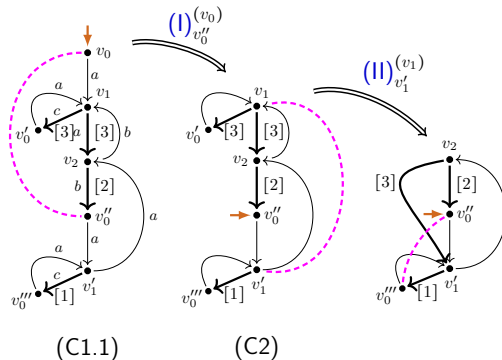


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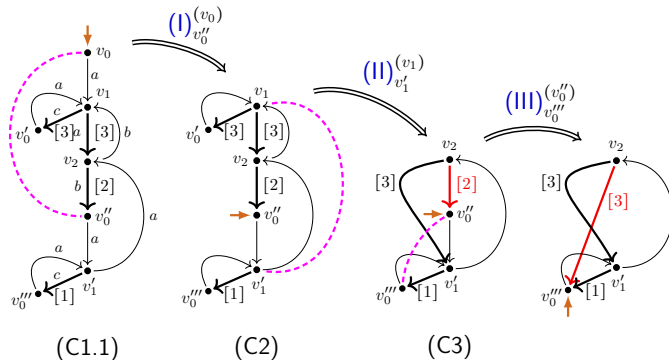


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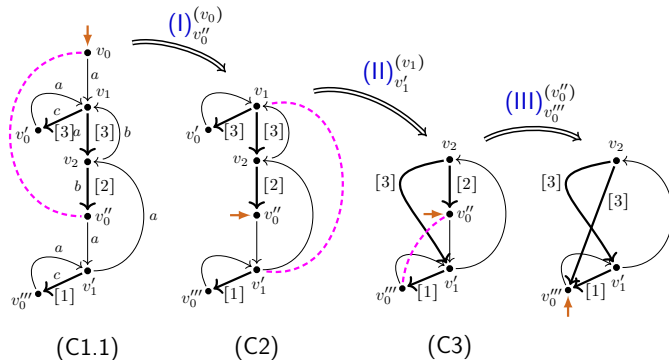


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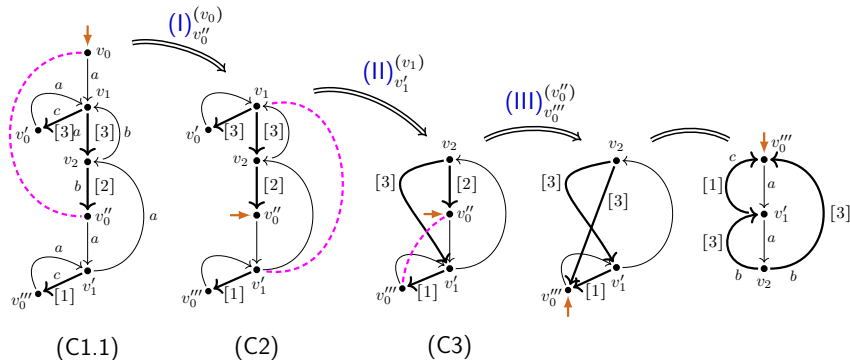


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Results facilitated by steps (st2)

Corollary ($\Leftarrow$ Fokkink-G, LICS 2020)

For every process graph G , TFAE:

- (i) G is $\llbracket \cdot \rrbracket_P$ -expressible by a $(*/\perp)$ regular expression
- (ii) The bisimulation collapse of G satisfies LLEE.
- (iii) The bisim. collapse of G is $\llbracket \cdot \rrbracket_P$ -expressible by a $(*/\perp)$ reg. expression.

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Lemma (G , IWC 2026)

LLEE, LLEE $\in$ PTIME ($\leq$ cubic in graph size).

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Theorem (Fokkink–G, LICS 2020)

Mil restricted to $(*/\perp)$ regular expressions is complete.

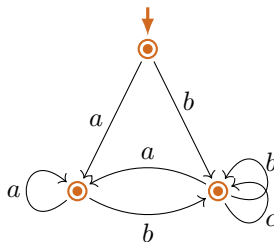
BBP (Bergstra, Bethke, Ponse)

Small steps for hard questions

- ▶ Regular expressions (also: 1-free/under-star-1-free $(*/\perp)$ expr.s)
- ▶ Milner's process interpretation P /semantics $\llbracket \cdot \rrbracket_P$
 - ▶ P -/ $\llbracket \cdot \rrbracket_P$ -expressible graphs ($\leadsto$ expressibility questions)
 - ▶ axioms for $\llbracket \cdot \rrbracket_P$ -identity ($\leadsto$ completeness question)
- ▶ **Small steps** for **hard questions**
 - (st1) reduction steps for well-behaved graphs under bisimilarity
 - ▶ $\llbracket \cdot \rrbracket_P$ -expressibility is decidable
 - (st2) collapse steps for LLEE graphs under bisimilarity
 - ▶ completeness for $(*/\perp)$ expressions
 - (st3) collapse steps for LLEE 1-graphs under bisimilarity, **as far as possible**
 - ▶ counterexample to collapse, but leads to completeness
- ▶ Interpretation–readback correspondences
 - (irc) LLEE/1-LLEE graphs $\hat{=}$ image of $P^\bullet$ restricted to $(*/\perp)$ /all expr.s
 - ▶ where $P^\bullet$ a compact refinement of P
 - ▶ crucial for P -/ $\llbracket \cdot \rrbracket_P$ -expressibility, and completeness questions
- ▶ Outlook

1-Graphs with LLEE $\hat{=}$ graphs with 1-LLEE

graph

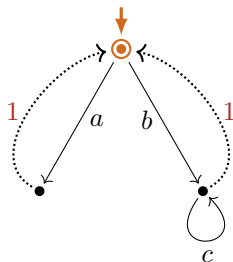


not LLEE

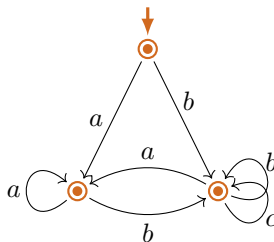
$$P((a \cdot 1 + b \cdot (c^* \cdot 1))^*)$$

1-Graphs with LLEE $\hat{=}$ graphs with 1-LLEE

1-graph



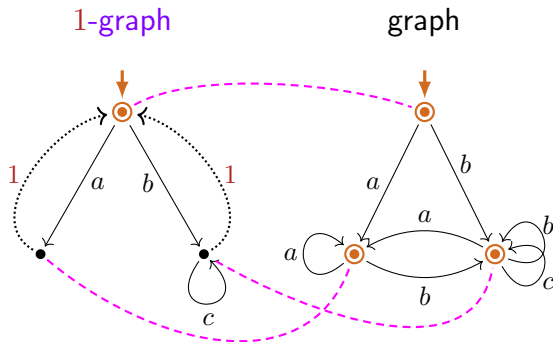
graph



$$\underline{P}((a \cdot 1 + b \cdot (c^* \cdot 1))^*)$$

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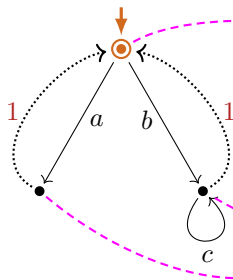
1-Graphs with LLEE $\hat{=}$ graphs with 1-LLEE



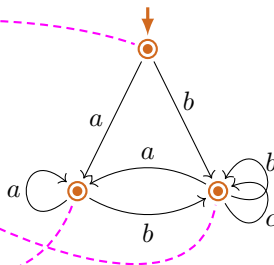
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1-Graphs with LLEE $\hat{=}$ graphs with 1-LLEE

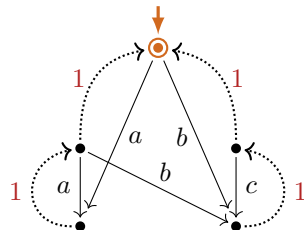
1-graph



graph



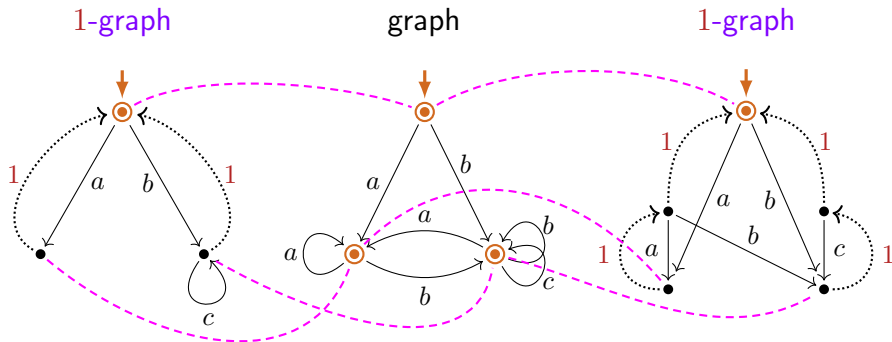
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$$\underline{P}((a \cdot 1 + b \cdot (c^* \cdot 1))^*) \quad P((a \cdot 1 + b \cdot (c^* \cdot 1))^*)$$

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1-Graphs with LLEE $\hat{=}$ graphs with 1-LLEE

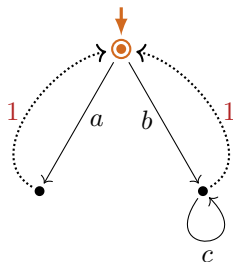


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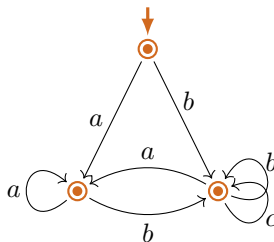
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1-Graphs with LLEE $\hat{=}$ graphs with 1-LLEE

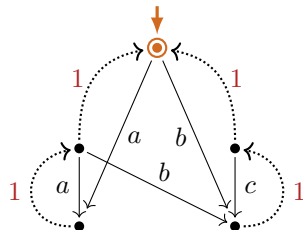
1-graph



graph



1-graph



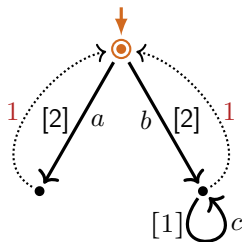
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1-Graphs with LLEE $\hat{=}$ graphs with 1-LLEE

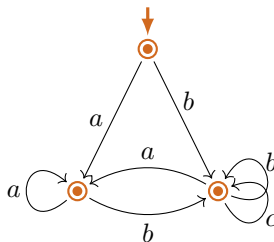
1-graph



LLEE

$$\underline{P}((a \cdot 1 + b \cdot (c^* \cdot 1))^*)$$

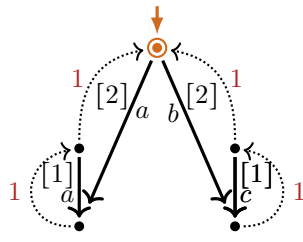
graph



not LLEE

$$P((a \cdot 1 + b \cdot (c^* \cdot 1))^*)$$

1-graph

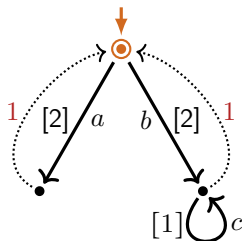


LLEE

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1-Graphs with LLEE $\hat{=}$ graphs with 1-LLEE

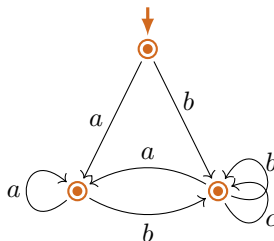
1-graph



LLEE

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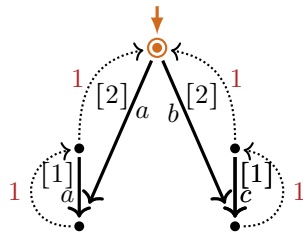
graph



1-LLEE

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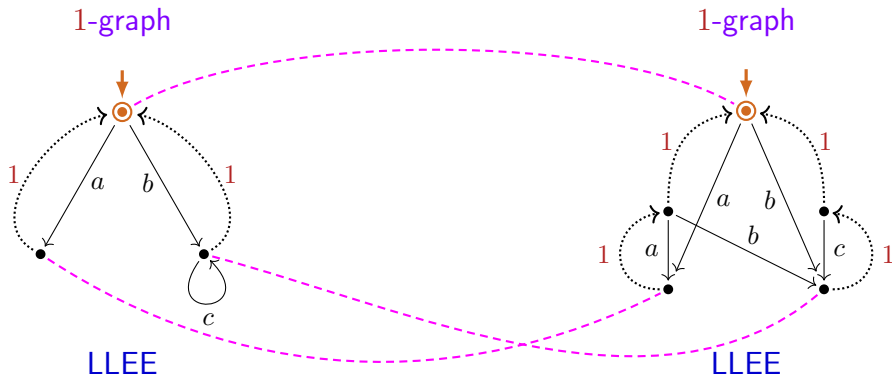
1-graph



LLEE

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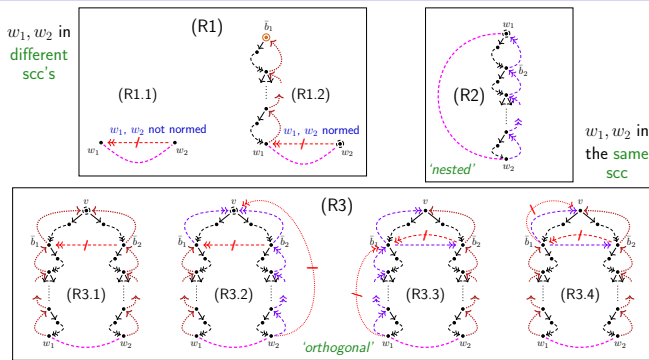


$$\underline{P}((a \cdot 1 + b \cdot (c^* \cdot 1))^*)$$



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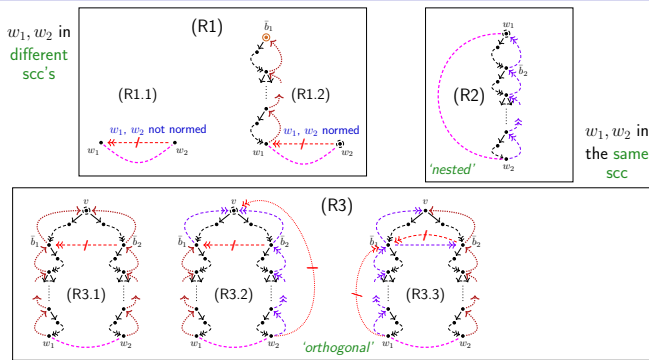
Reduced 1-bisimilarity redundancies in LLEE-1-charts



Lemma

Every *not collapsed* LLEE-1-chart contains 1-bisimilar vertices $w_1 \neq w_2$ (a *reduced 1-bisimilarity redundancy* $\langle w_1, w_2 \rangle$) of kind (R1), (R2), (R3).

Reduced 1-bisimilarity redundancies in LLEE-1-charts



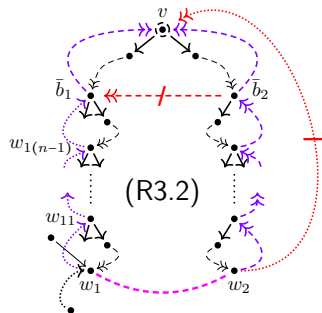
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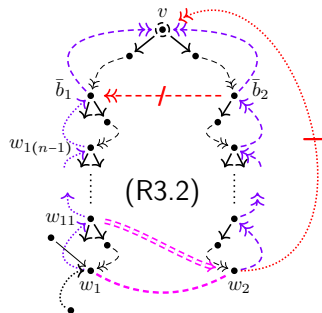
Lemma

Every *simple* reduced 1-bisimilarity redundancies in a LLEE-1-chart can be eliminated LLEE-preservingly.

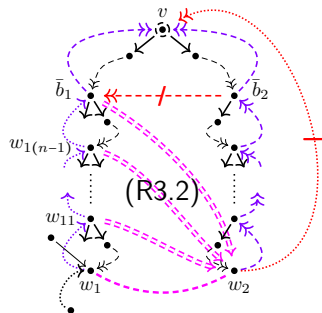
Elimination step of bisimilarity redundancies (R3.2)



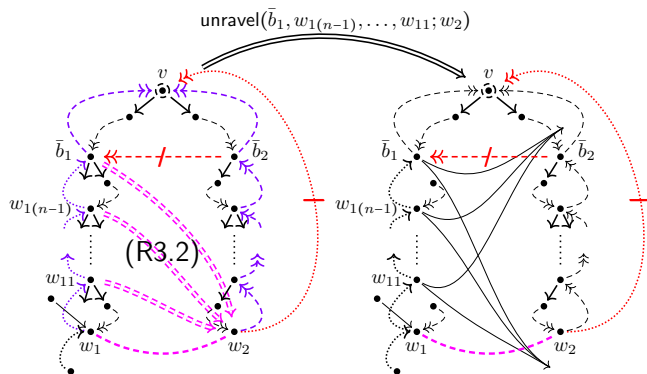
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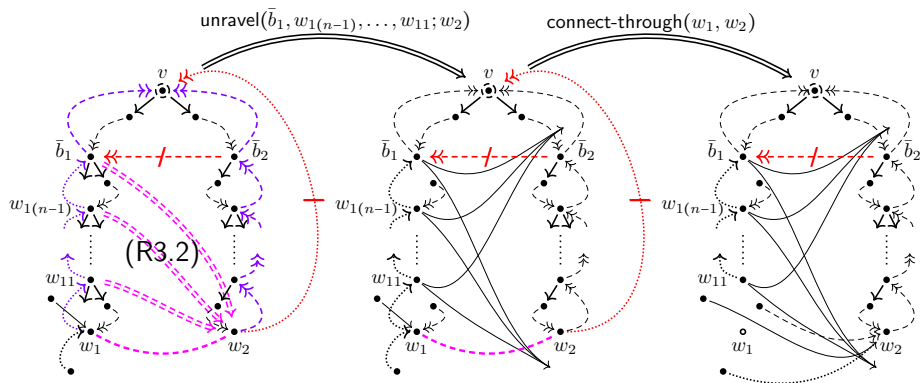
Elimination step of bisimilarity redundancies (R3.2)



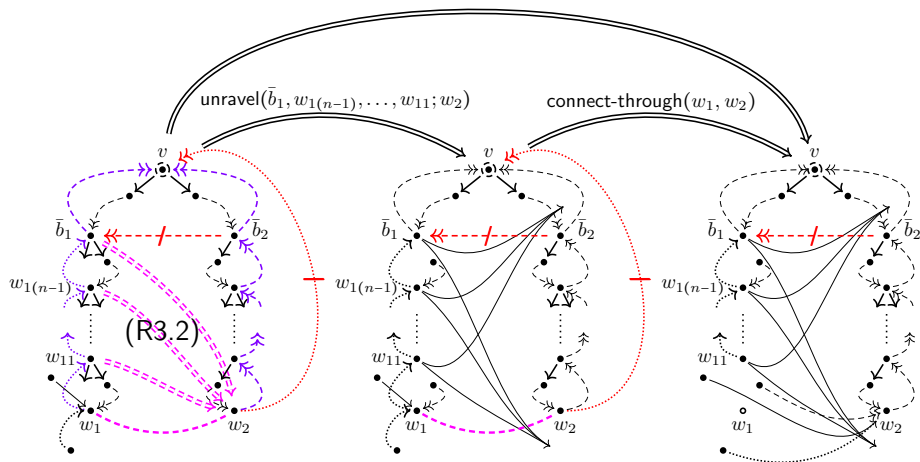
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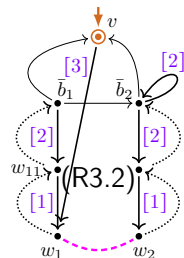
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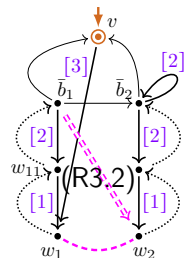
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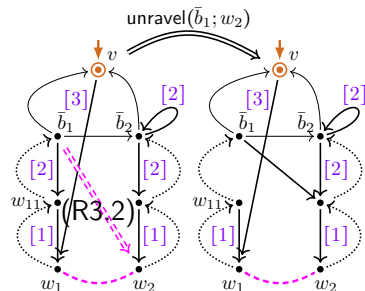
Elimination step of bisim. redundancy (R3.2) (example)



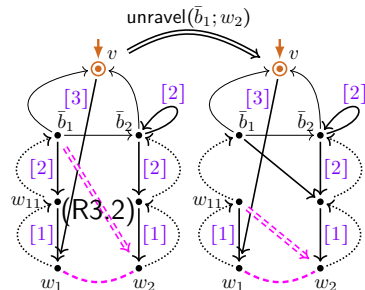
Elimination step of bisim. redundancy (R3.2) (example)



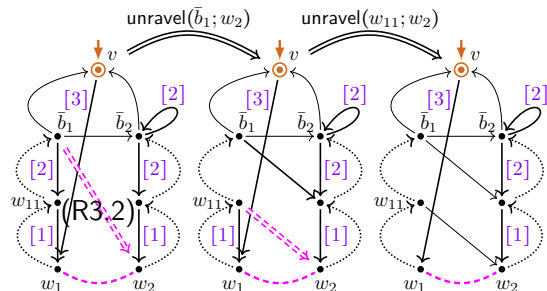
Elimination step of bisim. redundancy (R3.2) (example)



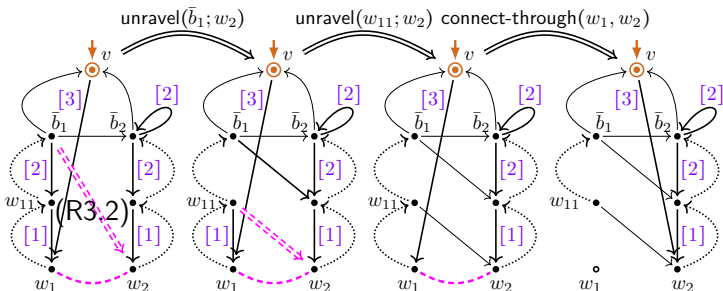
Elimination step of bisim. redundancy (R3.2) (example)



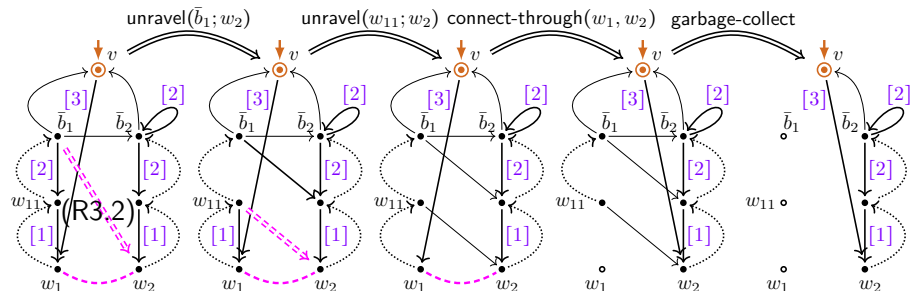
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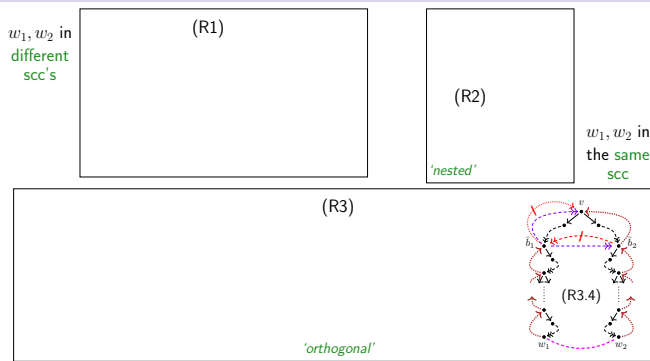
Elimination step of bisim. redundancy (R3.2) (example)



Elimination step of bisim. redundancy (R3.2) (example)



Reduced 1-bisimilarity redundancies in LLEE-1-charts



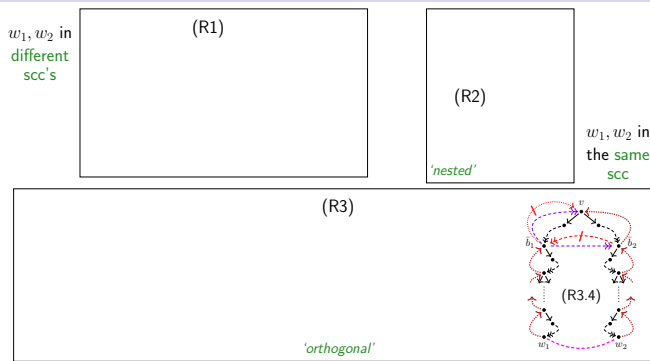
Lemma

Every *not collapsed* LLEE-1-chart contains 1-bisimilar vertices $w_1 \neq w_2$ (a *reduced 1-bisimilarity redundancy* $\langle w_1, w_2 \rangle$) of kind (R1), (R2), (R3).

Stumbling Block

How to LLEE-preservingly eliminate reduced 1-bisimilarity redundancies of kind (R3.4)?

Reduced 1-bisimilarity redundancies in LLEE-1-charts



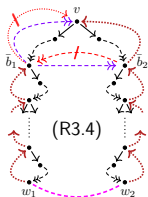
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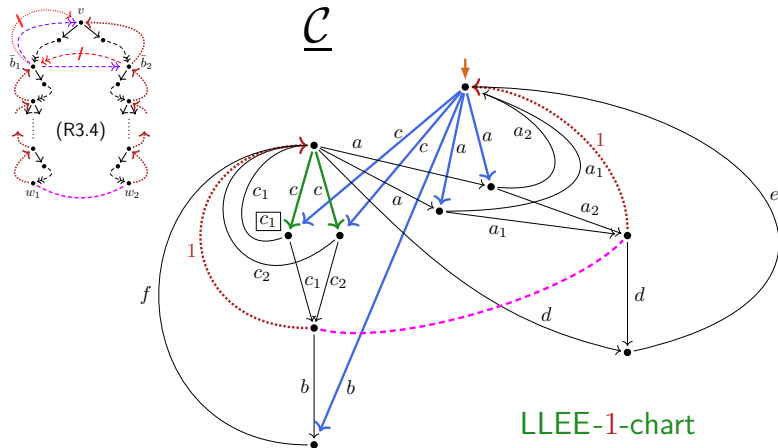
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Counterexample LLEE-preserving collapse

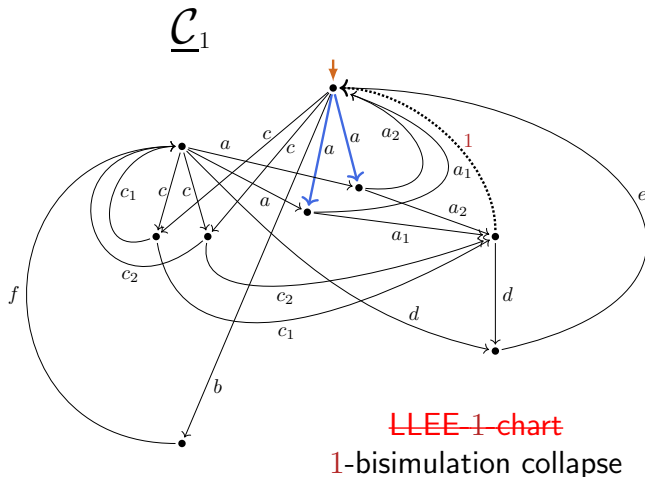


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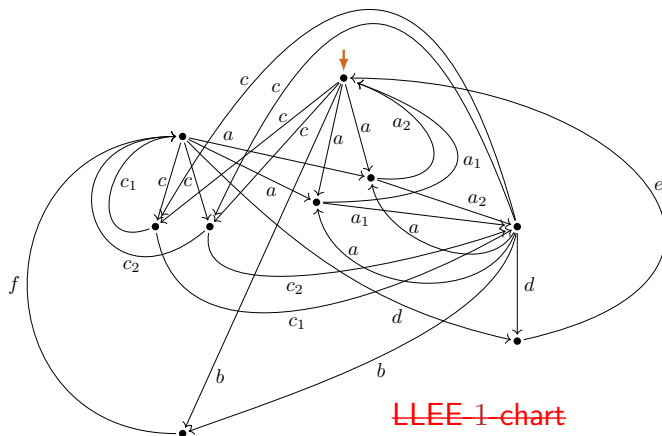


$\langle w_1, w_2 \rangle$ is a reduced
1-bisimilarity redundancy of kind (R3.4)

Counterexample LLEE-preserving collapse

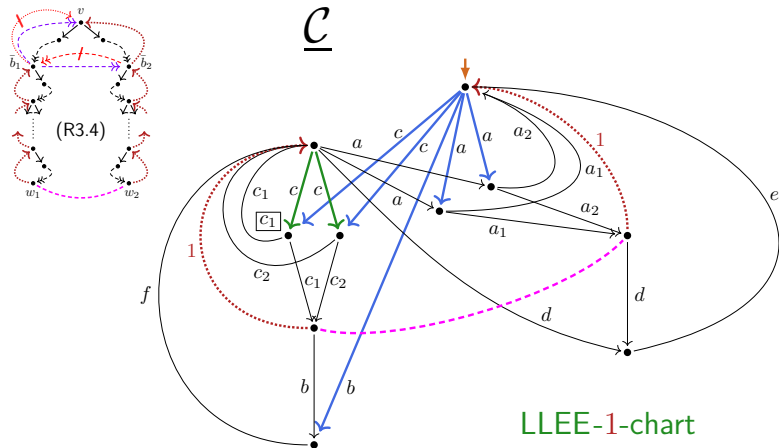


Counterexample LLEE-preserving collapse



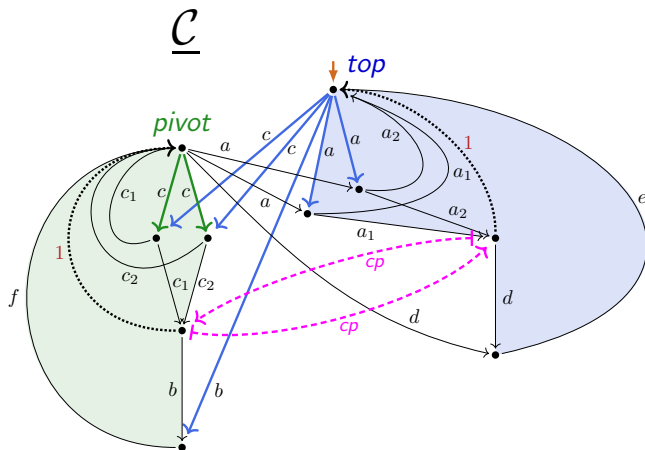
~~LLEE 1-chart~~
bisimulation collapse

Counterexample LLEE-preserving collapse

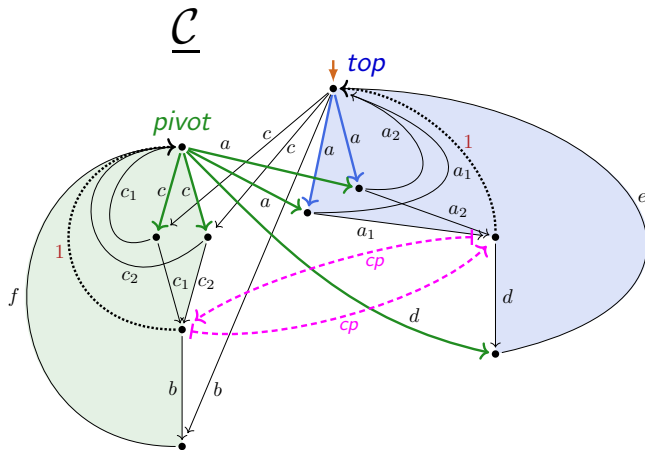


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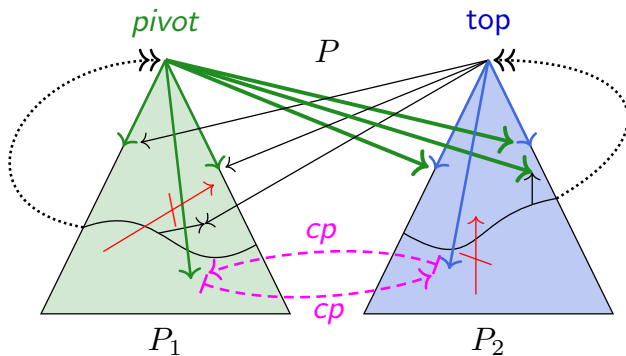
Twin-Crystal



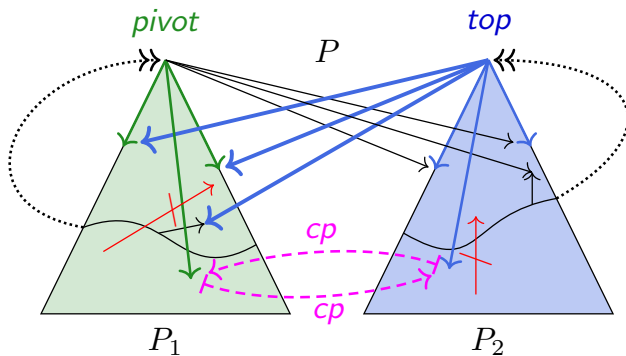
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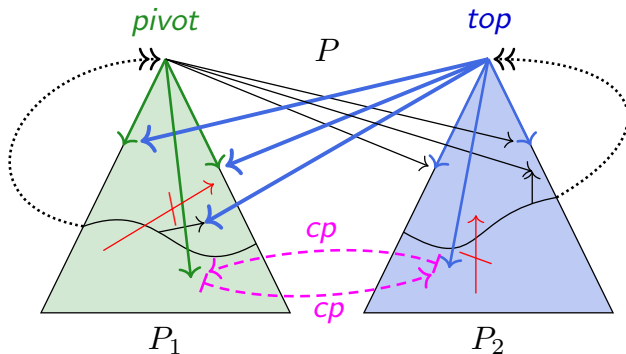
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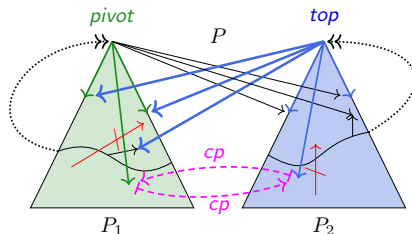


Twin-Crystal



- Provable solutions of **twin-crystals** are **complete**:
they can be transferred to their **bisimulation collapses**

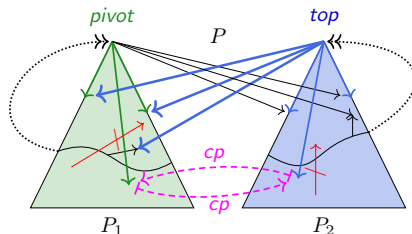
Crystallization



twin-crystal

Crystallized 1-charts = LLEE-1-charts that are collapsed apart from some scc's that are twin-crystals.

Crystallization

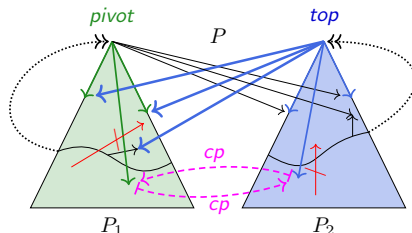


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(CR) Crystallization: Every LLEE-1-chart can be reduced under bisimilarity to a bisimilar crystallized 1-chart.

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twin-crystal

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(CR) Crystallization: Every LLEE-1-chart can be reduced under bisimilarity to a bisimilar crystallized 1-chart.

(CC) Every provable solution of a crystallized 1-chart gives rise to provable solution on the bisimulation collapse.

Results facilitated by steps (st3)

Proposition

The property 1-LLEE is not preserved under bisimulation collapse.

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Crystallisation Lemma for LLEE-1-graphs

Every 1-graph $\underline{G}$ with LLEE can be pseudo-collapsed/crystallized by elimination steps for bisimilarity redundancies of kinds (R1), (R2), (R3.1), (R3.2), (R3.3) (and related steps) to a 1-graph $\underline{G}_0$ such that:

- ▶ $\underline{G}_0 \leftrightarrow \underline{G}$,
- ▶ $\underline{G}_0$ is crystallised: all strongly connected components are collapsed or twin-crystals.

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Crystallisation facilitates to **adapt** the **completeness proof** for **BBP** (that uses preservation of LLEE under collapse) via a **crystallisation detour**.

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Crystallisation Lemma for LLEE-1-graphs

Every 1-graph $\underline{G}$ with LLEE can be **pseudo-collapsed/crystallized** by **elimination steps** for bisimilarity redundancies of kinds (R1), (R2), (R3.1), (R3.2), (R3.3) (and related steps) to a 1-graph $\underline{G}_0$ such that:

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Theorem (G , LICS 2022)

Milner's proof system **Mil** is **complete**.

Interpretation-readback correspondences

- ▶ Regular expressions (also: 1-free/under-star-1-free $(*/\perp)$ expr.s)
- ▶ Milner's process interpretation P /semantics $\llbracket \cdot \rrbracket_P$
 - ▶ P -/ $\llbracket \cdot \rrbracket_P$ -expressible graphs ($\leadsto$ expressibility questions)
 - ▶ axioms for $\llbracket \cdot \rrbracket_P$ -identity ($\leadsto$ completeness question)
- ▶ Small steps for hard questions
 - (st1) reduction steps for well-behaved graphs under bisimilarity
 - ▶ $\llbracket \cdot \rrbracket_P$ -expressibility is decidable
 - (st2) collapse steps for LLEE graphs under bisimilarity
 - ▶ completeness for $(*/\perp)$ expressions
 - (st3) collapse steps for LLEE 1-graphs under bisimilarity, as far as possible
 - ▶ counterexample to collapse, but leads to completeness
- ▶ Interpretation-readback correspondences
 - (irc) LLEE/1-LLEE graphs $\hat{=}$ image of $P^\bullet$ restricted to $(*/\perp)$ /all expr.s
 - ▶ where $P^\bullet$ a compact refinement of P
 - ▶ crucial for P -/ $\llbracket \cdot \rrbracket_P$ -expressibility, and completeness questions
- ▶ Outlook

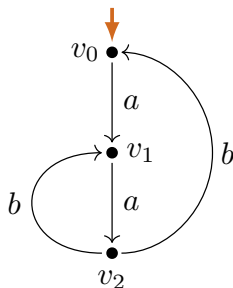
Readback

Lemma ($\sim G$, Fokkink, LICS 2020)

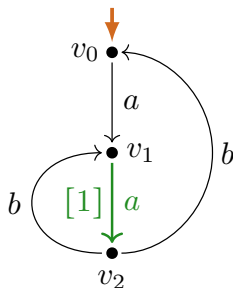
Process graphs G with LLEE are $\llbracket \cdot \rrbracket_P$ -expressible:

$$\text{LLEE}(G) \implies \exists e \in \text{RExp} \left(G \rightleftharpoons P(e) \right).$$

Readback from LLEE-witness (example)

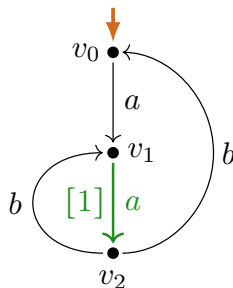


Readback from LLEE-witness (example)



layered
LEE-witness

Readback from LLEE-witness (example)



layered
LEE-witness

$$\begin{aligned} s(v_0) &= 0^* \cdot a \cdot s(v_1) \\ &=_{\text{Mil}^-} a \cdot s(v_1) \\ &=_{\text{Mil}^-} a \cdot (a \cdot (b + b \cdot a))^* \cdot 0 \end{aligned}$$

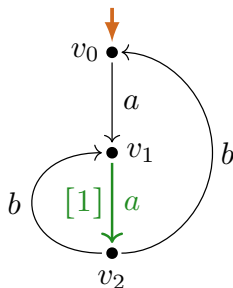
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$$\begin{aligned} s(v_1, v_1) &= 1 \\ s(v_0, v_1) &= 0^* \cdot a \cdot s(v_1, v_1) \\ &= 0^* \cdot a \cdot 1 \\ &=_{\text{Mil}^-} a \end{aligned}$$

Readback from layered LEE-witness (example)

$$s(v_0) = 0^* \cdot a \cdot s(v_1)$$

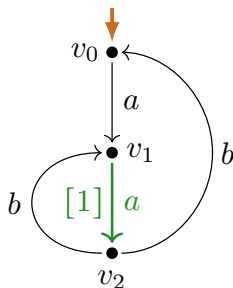


layered
LEE-witness

Readback from layered LEE-witness (example)

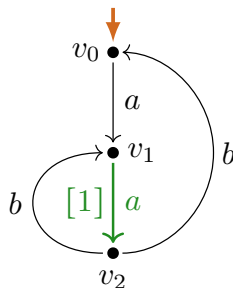
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layered
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Readback from layered LEE-witness (example)



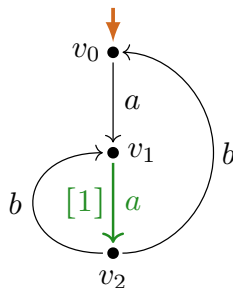
layered
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Readback from layered LEE-witness (example)



layered
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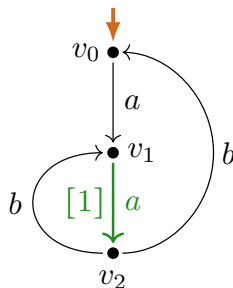
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layered
LEE-witness

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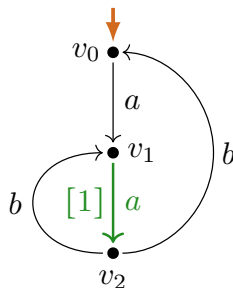
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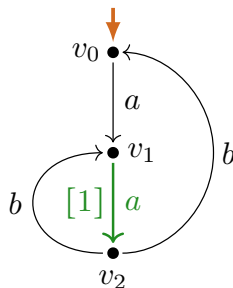
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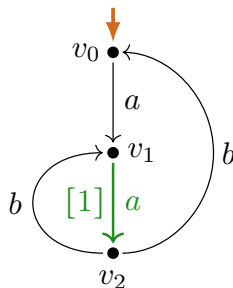
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Readback from layered LEE-witness (example)



layered
LEE-witness

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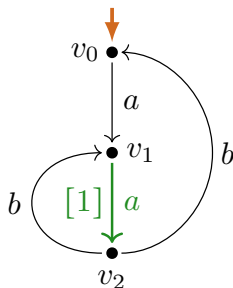
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Readback from layered LEE-witness (example)



layered
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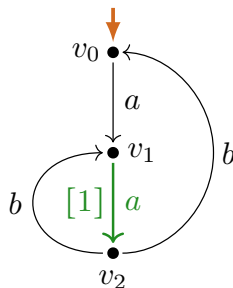
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Readback from layered LEE-witness (example)



layered
LEE-witness

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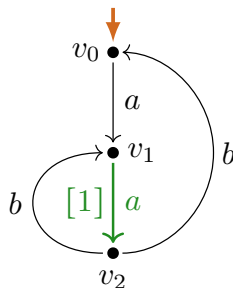
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Readback from layered LEE-witness (example)



layered
LEE-witness

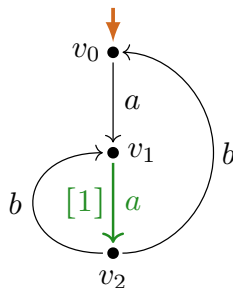
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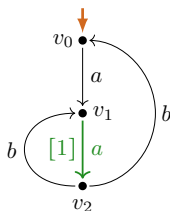
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$$\begin{aligned} s(v_1, v_1) &= 1 \\ s(v_0, v_1) &= 0^* \cdot a \cdot s(v_1, v_1) \\ &= 0^* \cdot a \cdot 1 \\ &=_{\text{Mil}^-} a \end{aligned}$$

Readback

Readback steps reverse $P^\bullet$ -defined transitions up to $=_{\text{Mil}^-}$ and $\leftrightarrow$.



layered
LEE-witness

$$\begin{aligned}
 s(v_0) &= 0^* \cdot a \cdot s(v_1) \\
 &=_{\text{Mil}^-} a \cdot s(v_1) \\
 &=_{\text{Mil}^-} a \cdot (a \cdot (b + b \cdot a))^* \cdot 0 \\
 s(v_1) &= (a \cdot s(v_2, v_1))^* \cdot 0 \\
 &=_{\text{Mil}^-} (a \cdot (b + b \cdot a))^* \cdot 0 \\
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 &= 0^* \cdot a \cdot 1 \\
 &=_{\text{Mil}^-} a
 \end{aligned}$$

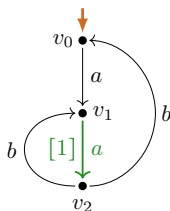
Readback

Lemma ($\sim G$, Fokkink, 2020)

Process graphs with LLEE are $\llbracket \cdot \rrbracket_P$ -expressible by $(*/\perp)$ reg. expressions:

$$\text{LLEE}(G) \implies \exists uf \in \text{RExp}^{(*/\perp)} (G \Leftrightarrow P(uf)).$$

Proof: Readback steps reverse $P^\bullet$ -defined transitions up to $=_{\text{Mil}^-}$ and $\Leftrightarrow$.



layered
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$$\begin{aligned} s(v_0) &= 0^* \cdot a \cdot s(v_1) \\ &=_{\text{Mil}^-} a \cdot s(v_1) \\ &=_{\text{Mil}^-} a \cdot (a \cdot (b + b \cdot a))^* \cdot 0 \\ s(v_1) &= (a \cdot s(v_2, v_1))^* \cdot 0 \\ &=_{\text{Mil}^-} (a \cdot (b + b \cdot a))^* \cdot 0 \\ s(v_2, v_1) &= 0^* \cdot (b \cdot s(v_1, v_1) + b \cdot s(v_0, v_1)) \\ &=_{\text{Mil}^-} 0^* \cdot (b \cdot 1 + b \cdot a) \\ &=_{\text{Mil}^-} b + b \cdot a \\ s(v_1, v_1) &= 1 \\ s(v_0, v_1) &= 0^* \cdot a \cdot s(v_1, v_1) \\ &= 0^* \cdot a \cdot 1 \\ &=_{\text{Mil}^-} a \end{aligned}$$

Readback (refined)

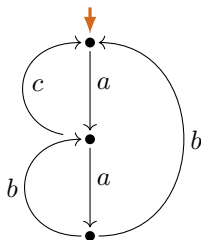
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Proof: Refine readback steps to precisely reverse $P^\bullet$ -defined transitions!

G_2



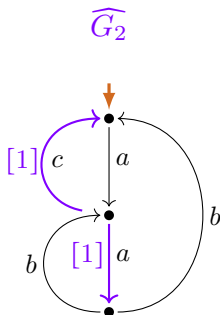
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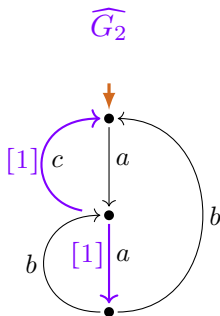
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$$P^\bullet(1 \cdot (\quad)^* \cdot 0)$$

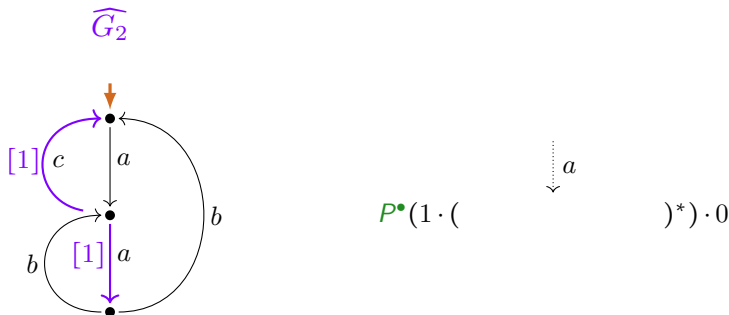
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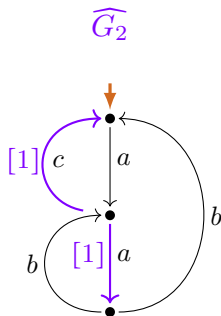
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$$P^\bullet(((1 \cdot a) \cdot (\quad)^*) \cdot 0)$$

$$P^\bullet(1 \cdot (\quad)^*) \cdot 0$$

$\downarrow a$

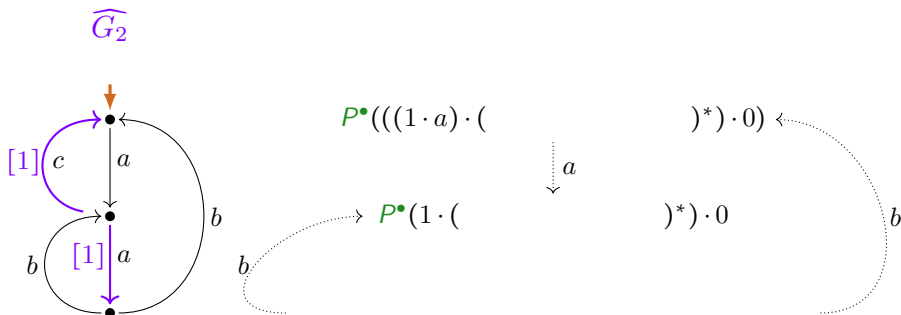
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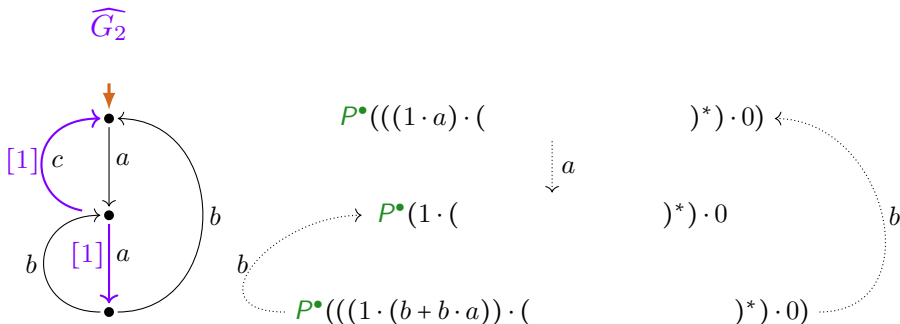
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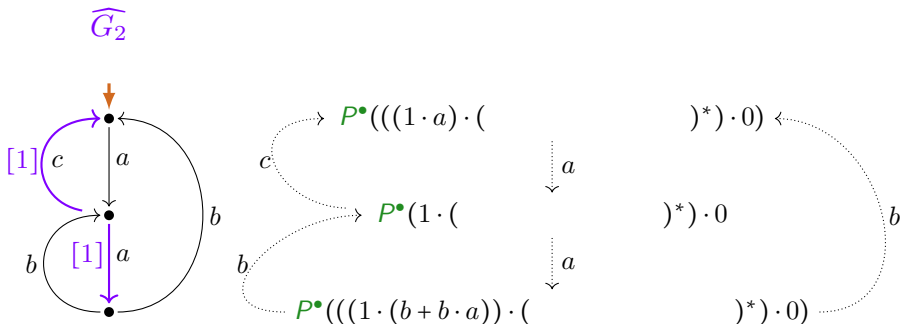
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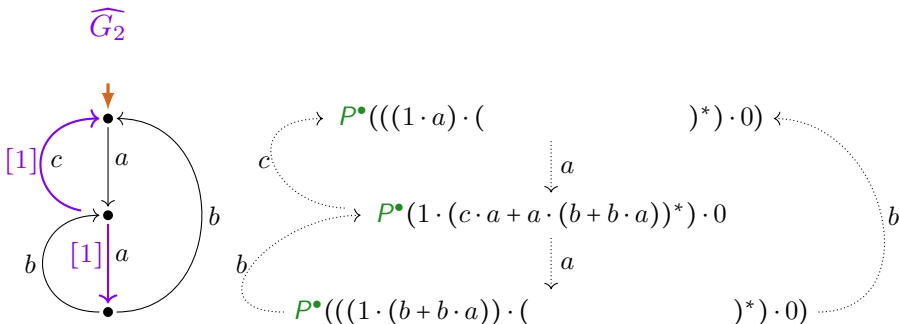
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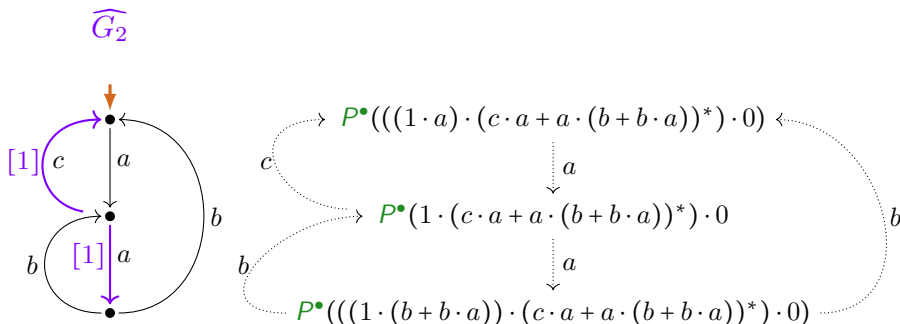
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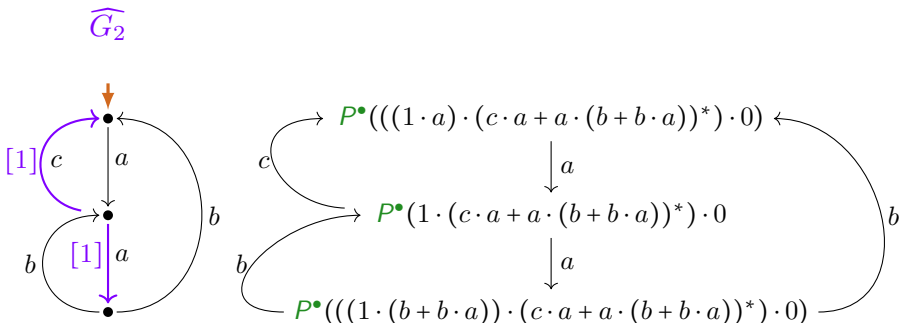
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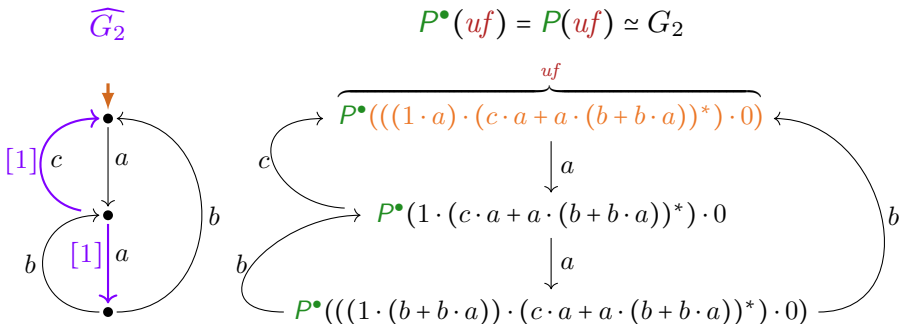
Readback (refined)

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Proof: Refine readback steps to precisely reverse $P^\bullet$ -defined transitions!



Summary

- ▶ Milner's process interpretation P /semantics $\llbracket \cdot \rrbracket_P$
 - ▶ P -/ $\llbracket \cdot \rrbracket_P$ -expressible graphs ($\leadsto$ expressibility question)
 - ▶ axioms for $\llbracket \cdot \rrbracket_P$ -identity ($\leadsto$ completeness question)
- ▶ Loop existence and elimination (LEE), layered form (LLEE)
- ▶ Small steps for hard questions
 - ▶ reduction steps for well-behaved specifications under bisimilarity
 - ▶ expressibility is decidable
 - ▶ collapse steps for LLEE graphs under bisimilarity
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 - ▶ collapse steps, as far as possible, for 1-LLEE graphs under bisimilarity
 - ▶ counterexample to collapse, but leads to completeness
- ▶ Interpretation–readback correspondences
 - ▶ LEE/1-LLEE characterise restricted to $(*/\perp)$ /unrestricted image of $P^\bullet$
 - ▶ where $P^\bullet$ a sharing-increased compact refinement of P

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Next aims, and refined questions

Expressibility problems (for $P^\bullet$ and $\llbracket \cdot \rrbracket_P$)

A1: graph structure of regular expression processes (LEE/1-LEE)

- ▶ $LLEE \hat{=} \text{im}(P^\bullet)((*/\perp))$ $1\text{-}LLEE \hat{=} \text{im}(P^\bullet)$ (TERMGRAPH'26)
- ▶ $LLEE \hat{=} \text{co-reducible}^+$ graphs ($\Leftarrow$ Laarakker/Kappè, TERMGRAPH'26)

A2: LEE/ $P^\bullet$ -Expressibility is decidable in $\leq$ cubic time.

$\Leftarrow$ confluence of loop elimination with pruning steps (IWC'26),
which also entails $LLEE = LEE$.

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Completeness problem, solution:

A3: motivation of crystallization

A4: details of crystallization procedure,
and completeness of Milner's proof system

TERMGRAPH / IWC Talks

TERMGRAPH (tomorrow, 19 July)

CG: *Towards a Characterisation of the Graph Structure of Process Interpretations of Regular Expressions*

Ext. Abstract <https://clegra.github.io/lf/pi-struct-TG-26.pdf>

- ▶ LEE $\hat{=}$ image of $P^\bullet|_{RExp(*/\pm)}$
- ▶ 1-LLEE $\hat{=}$ image of $P^\bullet$

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- ▶ $\text{LEE} \triangleq \text{image of } P^\bullet \big|_{\text{RExp}(* / \pm)}$
- ▶ $1\text{-LEE} \triangleq \text{image of } P^\bullet$

IWC (Friday, 24 July)

CG: *Loop Elimination in Process Graphs is Confluent When Pruning Steps are Added*

Ext. Abstract <https://clegra.github.io/lf/le-conf-IWC-26.pdf>

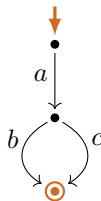
- ▶ critical-pair like analysis of loop elimination steps $\longrightarrow_{\text{elim}}$
- ▶ $\longrightarrow_{\text{elim}}$ forks are joinable by $\longrightarrow_{\text{elim}}$ steps and prune steps $\longrightarrow_{\text{prune}}$

Resources

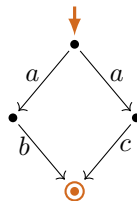
- ▶ Slides/abstract on clegra.github.io
 - ▶ slides: [.../1f/IFIP-1_6-2024.pdf](#)
 - ▶ abstract: [.../1f/abstract-IFIP-1_6-2026.pdf](#)
- ▶ CG: Closing the Image of the Process Interpretation of 1-Free Regular Expressions Under Bisimulation Collapse
 - ▶ TERMGRAPH 2024, [extended abstract](#)
- ▶ CG: The Image of the Process Interpretation of Regular Expressions is Not Closed under Bisimulation Collapse
 - ▶ [arXiv:2303.08553](#), 2021/2023.
- ▶ CG: Milner's Proof System for Regular Expressions Modulo Bisimilarity is Complete
 - ▶ LICS 2022, [arXiv:2209.12188](#), [poster](#)
- ▶ CG, Wan Fokkink: A Complete Proof System for 1-Free Regular Expressions Modulo Bisimilarity
 - ▶ LICS 2020, [arXiv:2004.12740](#), [video on youtube](#)
- ▶ CG: Modeling Terms by Graphs with Structure Constraints
 - ▶ TERMGRAPH 2018, [EPTCS 288](#), [arXiv:1902.02010](#)

Process semantics equality $=_{\llbracket \cdot \rrbracket_P}$

- Fewer identities hold for $=_{\llbracket \cdot \rrbracket_P}$ than for $=_L$:



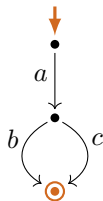
$$P(a \cdot (b + c))$$



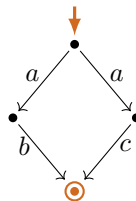
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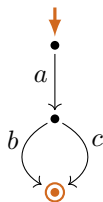
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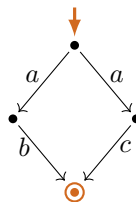
Process semantics equality $=_{\llbracket \cdot \rrbracket_P}$

- Fewer identities hold for $=_{\llbracket \cdot \rrbracket_P}$ than for $=_L$:



$$a \cdot (b + c)$$

$\neq_{\llbracket \cdot \rrbracket_P}$

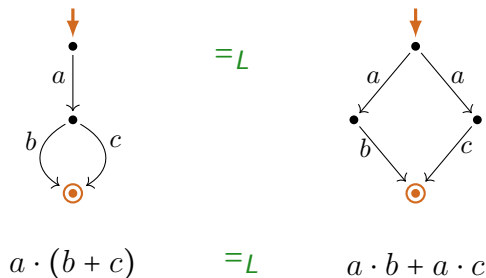


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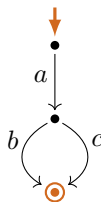
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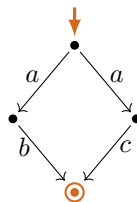
Process semantics equality $=_{\llbracket \cdot \rrbracket_P}$

- Fewer identities hold for $=_{\llbracket \cdot \rrbracket_P}$ than for $=_L$: $=_{\llbracket \cdot \rrbracket_P} \not\subseteq =_L$.



$$a \cdot (b + c)$$

$\neq_{\llbracket \cdot \rrbracket_P}$



$$a \cdot b + a \cdot c$$

$\neq_{\llbracket \cdot \rrbracket_P}$

Milner's proof system **Mil**

Axioms:

$$(A1) \quad e + (f + g) = (e + f) + g$$

$$(A2) \quad e + 0 = e$$

$$(A3) \quad e + f = f + e$$

$$(A4) \quad e + e = e$$

$$(A5) \quad e \cdot (f \cdot g) = (e \cdot f) \cdot g$$

$$(A6) \quad (e + f) \cdot g = e \cdot g + f \cdot g$$

$$(A7) \quad e = 1 \cdot e$$

$$(A8) \quad e = e \cdot 1$$

$$(A9) \quad 0 = 0 \cdot e$$

$$(A10) \quad e^* = 1 + e \cdot e^*$$

$$(A11) \quad e^* = (1 + e)^*$$

$$\text{But: } e \cdot (f + g) \neq e \cdot f + e \cdot g$$

$$\text{But: } e \cdot 0 \neq 0$$

Inference rules: rules of equational logic *plus*

$$\frac{e = f \cdot e + g}{e = f^* \cdot g} \text{RSP}^* \text{ (if } f \text{ does not terminate immediately)}$$

Milner's Completeness Question

Is **Mil** complete with respect to $=_{[\cdot]_P}$? (Does $e =_{[\cdot]_P} f \implies e =_{\text{Mil}} f$ hold?)

Milner's questions

(Q1) Completeness Question:

Is the proof system Mil complete for $=_{\llbracket \cdot \rrbracket_P}$?

(Q2) $\llbracket \cdot \rrbracket_P$ -Expressibility Question:

What structural property characterizes process graphs that are $\llbracket \cdot \rrbracket_P$ -expressible?

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What *structural property* characterizes
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- ▶ is decidable (Baeten/Corradini/G, 2007)

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- ▶ partial new answer (G/Fokkink, 2020):
 - ▶ bisimulation collapse has *loop existence & elimination property* (LEE)
if expressible by *under-star-1-free* regular expression

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- ▶ series of partial completeness results for:
 - ▶ exitless iterations (Fokkink, 1998)
 - ▶ with a stronger fixed-point rule (G, 2006)
 - ▶ under-star 1-free, and without 0 (Corradini/de Nicola/Labella, 2004)
 - ▶ with 0 but under-star-1-free (G/Fokkink, 2020)

(Q2) $\llbracket \cdot \rrbracket_P$ -Expressibility Question:

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Is the proof system *Mil* *complete* for $=_{\llbracket \cdot \rrbracket_P}$?

- ▶ Yes! (G, 2022, proof summary, employing LEE and crystallization)
- ▶ series of partial completeness results for:
 - ▶ exitless iterations (Fokkink, 1998)
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Question (Q2) specialized

(Q2)₀ P -Expressibility and P - $(*/\perp)$ -Expressibility:

What *structural property* characterizes:

- ▶ process graphs that are P -expressible ?
(... in the *image of P* ?)
- ▶ process graphs that are P -expressible by $(*/\perp)$ regular expressions?
(... in the *image of $(*/\perp)$ expressions under P* ?)